THE IRREDUCIBLE UNITARY REPRESENTATIONS OF UNIVERSAL BURGER-MOZES GROUPS IN INFINITE ARITY

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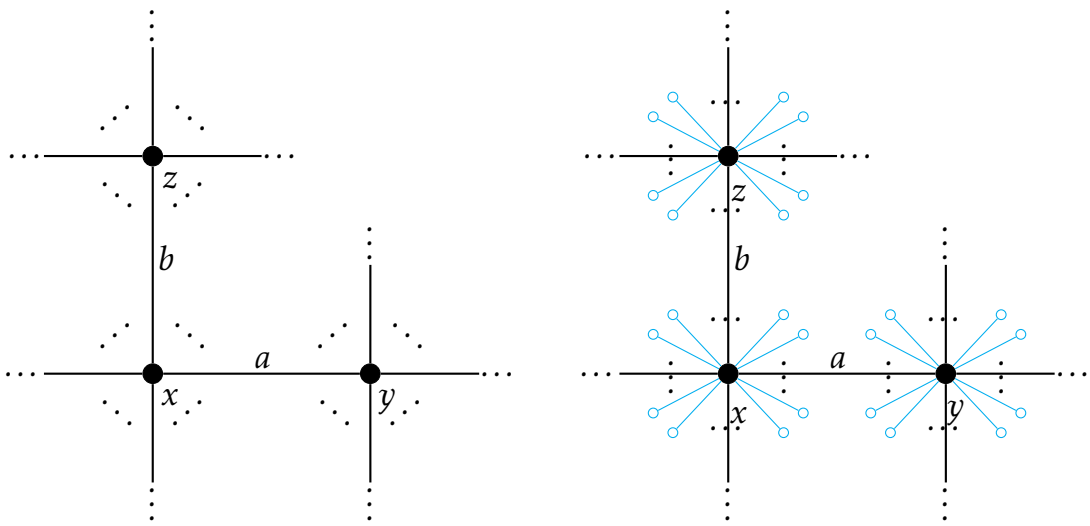
Abstract. Given a closed permutation group Γ over a countable set Ω and a suitable Ω-coloring of the regular tree of countably infinite arity $\mathbf{T}$, we define the group G consisting of all the automorphisms of $\mathbf{T}$ that act around each vertex of $\mathbf{T}$ as an element of Γ. This is a generalization of a construction carried out by Burger and Mozes in the finite arity case.

We fully classify the continuous irreducible unitary representations of G when $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. This is achieved by building on a similar classification by Ol'šanskiĭ for the group $\mathrm{Aut}(\mathbf{T})$, which is the special case where Γ is the group of all the bijections of Ω. This provides new examples of locally Roelcke-precompact Polish groups of Type I. We also prove that G has the Howe-Moore property under the same hypothesis on Γ.

Introduction

The notion of *Type I groups*, arising from the study of von Neumann algebras, plays a key role in abstract harmonic analysis. It forms a dividing line between topological groups whose representation theory is tractable and those for which it is too wild to be fully understood. We postpone the formal definition to the end of Section 2 and only give an illustration: by a celebrated theorem of Glimm [Gli61], a locally compact group G is of Type I if and only if every continuous unitary representation of G decomposes *uniquely* as a direct integral of irreducible representations. In any case, deciding if a given topological group is of Type I or not is a crucial step in the study of its representation theory.

For fundamental reasons, all abelian groups are of Type I, even as discrete groups. In fact, a discrete group is of Type I if and only if it is virtually abelian by a result of Thoma [Tho68]. In the non-discrete case, the picture is a lot richer while remaining, to this day, far less complete. The



(a) The regular tree $\mathbf{T}$ (b) The expansion $\mathbf{T}^{\mathrm{eq}}$

Figure 1. The trees $\mathbf{T}$ and $\mathbf{T}^{\mathrm{eq}}$

Haar measure being an invaluable tool in representation theory, the classical case is the locally compact one. For example, all compact groups are of Type I as well as all of the connected semi-simple or nilpotent Lie groups [Har54, Dix59, Kir61] and all connected real algebraic groups [Dix57]. On the other hand, the Mautner group is a solvable connected Lie group which is not of Type I (due to Mautner, unpublished, see [Bag78]).

At the opposite end from connected Lie groups are the totally disconnected locally compact groups, for which a large part of the landscape is occupied by those who act on trees. In 1999, Nebbia obtained a striking result by showing that, for such groups, having a very transitive action is a necessary condition for having a very tame representation theory [Neb99]. In the same paper, he conjectured that the converse also holds. Nebbia's conjecture remains open in its full generality but Houdayer and Raum [HR19] provided a positive answer for a particularly interesting class of groups: the *Burger-Mozes universal groups*. Given $\mathbf{T}_d$ the regular tree of finite arity $d \in \mathbb{N}_{\geqslant 3}$, one can color the unoriented edges of $\mathbf{T}_d$ with labels in $\{1,\dots,d\}$. Then, every pair (g,x) where g is an automorphism of $\mathbf{T}_d$ and x is a vertex of $\mathbf{T}_d$ induces a bijection of $\{1,\dots,d\}$ by keeping track of how g sends the edges around x to the edges around $g(x)$. This bijection is called the *local action* of g at x. Now, given a group $F \leqslant \mathrm{Sym}(\{1,\dots,d\})$, one can consider the subgroup $\mathbf{U}(F)$ of $\mathrm{Aut}(\mathbf{T}_d)$ consisting of all the automorphisms g of $\mathbf{T}_d$ such that for every vertex x in $\mathbf{T}_d$, the local action of g at x belongs F.

Theorem ([Ol'80, HR19])**.** *Let $\mathbf{T}_d$ be the regular tree of finite arity $d \in \mathbb{N}_{\geqslant 3}$ and let $F \leqslant \mathrm{Sym}(\{1,\dots,d\})$. The following properties are equivalent:*

(i) $\mathbf{U}(F)$ is of Type I,
(ii) the action $F \curvearrowright \{1,\dots d\}$ is 2-transitive.

In this paper, we are interested in analogous constructions and results in infinite arity. More precisely, we replace $\mathbf{T}_d$ with the regular tree $\mathbf{T}$ of countably infinite degree. This takes us out of the locally compact world but still leaves us in the Polish realm, that is of separable and completely metrizable groups.

Representations of Polish groups have been studied at least since the 1970's. One of the first examples of Type I non-locally compact groups is the infinite symmetric group $\mathfrak{S}_\infty$ of all permutations of a countably infinite set [Lie72]. In the years that followed, Ol'šanskiĭ developed the *semigroup* method and used it to find non locally-compact Type I groups by classifying all the unitary representations of a variety of groups such as the isometry group of the separable (real or complex) Hilbert space, the symplectic variations of it, as well as some infinite dimensional general linear groups over finite fields (see the survey [Ol'91]). More closely related to our work, Ol'šanskiĭ also classified the irreducible representations of $\mathrm{Aut}(\mathbf{T})$, as well as those of $\mathbf{U}(F)$ when $F \curvearrowright \{1,\dots,d\}$ is 2-transitive [Ol'80].

In Section 2.2, we introduce[1] the infinite arity analogue of universal Burger-Mozes groups. Indeed, one can carry out a similar construction as the one described above by replacing $\mathbf{T}_d$ with $\mathbf{T}$ and $F \curvearrowright \{1,\dots,d\}$ with $\Gamma \curvearrowright \Omega$ for some infinite permutation group Γ over a countable set Ω. We now fix some permutation group Ω and denote by G the subgroup of $\mathrm{Aut}(\mathbf{T})$ obtained in this manner. See Definition 1.1.

The goal of this paper is to generalize Ol'šanskiĭ's classification of the irreducible representations of $\mathrm{Aut}(\mathbf{T})$ to G under transitivity hypotheses on Γ. The core idea is that control over the representation theory of the *local action* $\Gamma \curvearrowright \Omega$ gives us control over the representation theory of the *global action* $G \curvearrowright \mathbf{T}$. While finite groups are all well-behaved in that regard, infinite arity allows for non-locally compact inputs to which the vast majority of the classical theory does not apply. It is thus essential to restrict one's attention to instances of the construction where the representation theory of the local action is well understood.

One important class of examples is that of oligomorphic permutation groups, that is the groups $\Gamma \leqslant \mathrm{Sym}(\Omega)$ such that for every $n \in \mathbb{N}$, the diagonal action $\Gamma \curvearrowright \Omega^n$ has only finitely many

[1] Another generalization of Burger-Mozes groups in infinite arity called *box products* has already been introduced [Smi17]. Using a different formalism, Smith studied the structure and dynamics on bi-regular trees and focuses on locally compact instances of this construction.

orbits. This can be seen as a relaxation of high transitivity and has topological consequences on G akin to compactness. In particular, Tsankov fully classified the unitary representations of these groups [Tsa12] in a way that reminds of the Peter-Weyl Theorem. A direct consequence of Tsankov's classification is that oligomorphic groups are of Type I. Another classification theorem in the same flavor was obtained in [BJJ25], adding new examples to the list of non-locally compact Type I groups which moreover have non-trivial large scale geometry.

A convenient way of thinking of permutation groups is through model theory. Indeed, countable first-order structures come equipped with their automorphism group and every permutation group appears as such. In this framework, oligomorphic permutation groups correspond exactly to automorphism groups of $\aleph_0$-categorical structures. Examples of oligomorphic groups thus include $\mathrm{Sym}(\Omega)$, the group $\mathrm{Aut}(\mathbb{Q},<)$ of order-preserving bijections of the rationals, or $\mathrm{Homeo}(\{0,1\}^{\mathbb{N}})$ seen as the automorphism group of countable atomless boolean algebra. A general method for building such structures is via Fraïssé theory: suitable classes of finite structures can be amalgamated into a uniquely determined universal countable object, called its *Fraïssé limit*, see e.g. [Hod93, Sec. 7.4]. For instance, the Fraïssé limit of finite orders is $(\mathbb{Q},<)$ and the Fraïssé limit of finite graphs, called the Rado graph, is also $\aleph_0$-categorical. However, oligomorphy alone will not be sufficient for our purposes. As hinted at by Houdayer and Raum's result above, our classification theorem will also require 2-transitivity of the local action. Fortunately, it is well understood what makes a Fraïssé limit $\aleph_0$-categorical/2-transitive or not. Examples where both properties hold include the separable infinite dimensional projective space $\mathbb{P}^\infty(q)$ on the finite field $\mathbb{F}_q$ or the $\aleph_0$-categorical hypergraphs from [EP02]. See the monograph [Cam90] or [Mac11] for detailed discussions on oligomorphic groups.

We will want to make use of the descriptive power of model theory and refer to results about permutation groups that are phrased in its language. To that purpose, we will assume that Ω is the underlying set of a countable structure $\mathfrak{M}$ and that $\Gamma = \mathrm{Aut}(\mathfrak{M})$. Moreover, we will identify $\mathfrak{M}$ with its underlying set Ω. If $\mathfrak{M}$ is relational, there is a natural first order structure on $\mathbf{T}$ whose automorphism group is precisely G but we will not use it. In fact, all the results and most of the proofs are about permutation groups, not model theory. We insist in particular on the fact that $\Gamma \curvearrowright \Omega$ is oligomorphic if and only if $\mathfrak{M}$ is $\aleph_0$-categorical.

The study of the irreducible unitary representations of G splits into two very distinct parts. In the classical terminology, a representation of G is called *spherical* if it admits a non-zero vector that is invariant under the action of a vertex stabilizer. Ol'šanskiĭ's classification of spherical irreducible representations of $\mathrm{Aut}(\mathbf{T})$ extends with the same proof to every 2-transitive subgroup of $\mathrm{Aut}(\mathbf{T})$ [Ol'80, Sec. 5.3–5.5]. In particular, it applies to G provided that $\Gamma \curvearrowright \mathfrak{M}$ is 2-transitive and this part of the unitary dual of G is independent of the choice of the 2-transitive permutation group Γ. We still give a brief description of these representations for the interested reader that is familiar with the GNS construction. Fix $x \in \mathbf{T}$, let $c \in [-1,1]$ and denote by d the graph distance in $\mathbf{T}$. The map $\varphi_c \colon G \longrightarrow \mathbb{C}$ defined by $\varphi_c(g) = c^{d(g(x),x)}$ is positive definite. Moreover, the representation π_c obtained from φ_c by the GNS construction is irreducible and spherical. In fact, every spherical irreducible representation of G is isomorphic to π_c for some unique $c \in [-1,1]$. It also follows from Ol'šanskiĭ's results that the process of restricting representations of $\mathrm{Aut}(\mathbf{T})$ to G induces a one-to-one correspondence between the spherical irreducible representations of $\mathrm{Aut}(\mathbf{T})$ and those of G.

The main result of this paper is a full description of the non-spherical representations of G, irreducible or not, provided that Γ is oligomorphic and 2-transitive. The core of it is contained in the following statement and a more precise description of the non-spherical irreducible representations of G will be given in Section 2, Theorem 2.13. Inputting $\Gamma = \mathrm{Sym}(\Omega)$, we recover Ol'šanskiĭ's classification. See Subsection 3.3 for reminders about commensurators.

Theorem 1. *Assume that $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive.*

1. *Every non-spherical unitary representation of G splits as a sum of irreducible subrepresentations.*

2. *Every non-spherical irreducible representation of G is of the form* $\mathrm{Ind}_{C(U)}^{G}(\sigma)$ *where $U \leqslant G$ is an open subgroup of G, $C(U)$ is its commensurator, $U \trianglelefteq$, and σ is an irreducible representation of the finite group $C(U)/U$.*

This theorem is very similar to Tsankov's [Tsa12, Th. 4.2]. However, the representation theory of G significantly differs from that of oligomorphic groups because of the spherical case. Indeed, oligomorphic groups only have countably many irreducible representations and all their representations split as sums of irreducibles. The same holds for G if one forbids sphericity but both parts fail in general. Recall also that oligomorphic groups have Property (T) [ET16, Th. 1.1] whereas G does not, not even in the weaker definition one can give using coarsely bounded subsets instead of compact subsets. We still deduce the following from our classification. The proof closely follows that of [Ol'80, Th. 5.7], see Corollary 2.15 below.

Corollary 1. *Assume that $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. Then G is of Type I.*

We note that the previous two results also apply to *dissociated* 2-transitive permutation groups in the sense of [BJJ25]. Indeed, this is precisely the assumptions one needs to carry out the proofs in [Ol'80, Sec. 5].

Finally, we also obtain the following geometrical result, proved in Section 3. It illustrates the fact that the large scale geometry of these group is well captured by its representation theory, after passing to the following index-2 simple subgroup:

$$G^{+} := \{g \in G \colon \forall x \in \mathbf{T},\ d(g(x), x) \text{ is even}\}.$$

Theorem 2. *Assume that $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. Then G has the Howe-Moore property.*

As an application, we give a description of all the open subgroups of G up to finite index in 3.9.

We emphasize on the fact that the proof of this result does not use the classification. However, combining the results of sections 2 and 3 provide a deep understanding of the open subgroups of G, see Corollary 3.9.

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1. Preliminaries

1.1. Permutation groups, colored trees and imaginaries.

Given Ω a countable set, $\mathrm{Sym}(\Omega)$ denotes the group of all bijections of Ω. We equip it with the *permutation group topology*, i.e. the Polish group topology given by pointwise convergence on Ω discrete. A *closed permutation group* over Ω is a closed subgroup of $G \subseteq \mathrm{Sym}(\Omega)$. Given $A \subseteq \Omega$, we will write G_A for the *pointwise stabilizer* of A in G and $G_{\{A\}}$ for the *setwise stabilizer* of A in G, that is:

$$G_A := \{g \in G \colon \forall a \in A,\ g(a) = a\} \qquad \& \qquad G_{\{A\}} = \{g \in G \colon g(A) = A\}.$$

Trees, i.e. acyclic connected graphs, will be identified with the set of their vertices. In particular, if X is a tree, "$x \in X$" will always mean that x is a vertex of X. The set of vertices of X *adjacent* to $x \in X$, called the *neighbors* of x, will be denoted $V_X(x)$. We will write $x \sim y$ if x and y are adjacent vertices of a tree. If x has exactly one neighbor in X, it will be called a *leaf* of X. We will denote by $E(X)$ the set of *unoriented edges* of X, i.e. the set of pairs $\{x, y\}$ of adjacent vertices in X, and we will write $E_X(x)$ for the set of all edges of X that involve x.

A *path* from x to y is a tuple $(x_0, \ldots, x_n)$ of vertices of X such that $x_0 = x$, $x_n = y$ and $x_i \sim x_{i+1}$ for every $i < n$. In this setting, $n \in \mathbb{N}$ is the *length* of this path. For every $x, y \in \mathbf{T}$, there exists a

unique *shortest path* $[x,y]$ from x to y and we define $d(x,y)$ to be the length of $[x,y]$. Then (X,d) is a metric space and $\mathrm{Aut}(X) = \mathrm{Iso}(X,d)$. We will write, for every $n \in \mathbb{N}$:

$$B_X(x,n) = \{z \in X \colon d(x,z) \leqslant n\} \qquad \text{and} \qquad S_X(x,n) = \{z \in X \colon d(x,z) = n\}.$$

For a subset or tuple $E \subseteq X$, $\langle E \rangle$ will denote the subtree of X generated by E, meaning:

$$\langle E \rangle = \bigcup \{[z,z'] \colon z,z' \in E\}.$$

$\mathbf{T}$ will denote the *regular tree with countably infinite arity*. Given a countably infinite first-order structure $\mathfrak{M}$, an $\mathfrak{M}$-*coloring* of $\mathbf{T}$ is a labeling of its unoriented edges by elements of $\mathfrak{M}$. More precisely, it is a map $\chi \colon E(\mathbf{T}) \longrightarrow \mathfrak{M}$ such that for every $x \in \mathbf{T}$, the restriction $\chi \colon E_{\mathbf{T}}(x) \longrightarrow \mathfrak{M}$ is a bijection. A back and forth argument shows that there exists a unique $\mathfrak{M}$-coloring of $\mathbf{T}$ up to conjugation by an element of $\mathrm{Aut}(\mathbf{T})$.

Given $g \in \mathrm{Aut}(\mathbf{T})$ and $x \in \mathbf{T}$, a coloring χ allows us to define the *local action* g_x *of* g *at* x. Indeed, let g_x be the unique bijection of $\mathfrak{M}$ such that the following diagram commutes:

$$\begin{tikzcd} E_{\mathbf{T}}(x) \arrow[r, "g"] \arrow[d, "\chi"'] & E_{\mathbf{T}}(g(x)) \arrow[d, "\chi"] \\ \mathfrak{M} \arrow[r, "g_x"] & \mathfrak{M} \end{tikzcd}$$

Note that the map $(g,x) \in \mathrm{Aut}(\mathbf{T}) \times \mathbf{T} \longmapsto g_x \in \mathrm{Sym}(\mathfrak{M})$ is a *cocycle*:

$$\forall g,h \in \mathrm{Aut}(\mathbf{T}), \forall x \in \mathbf{T},\ (gh)_x = g_{h(x)} \circ h_x\ .$$

Definition 1.1. Let $\mathfrak{M}$ be a countably infinite structure and let χ be an $\mathfrak{M}$-coloring of $\mathbf{T}$. The *Burger-Mozes universal group with prescribed local action* $\mathrm{Aut}(\mathfrak{M}) \curvearrowright \mathfrak{M}$ is the following closed subgroup of $\mathrm{Aut}(\mathbf{T})$:

$$\mathrm{Aut}_{\mathfrak{M}}(\mathbf{T}) := \{g \in \mathrm{Aut}(\mathbf{T}) \colon \forall x \in \mathbf{T},\ g_x \in \mathrm{Aut}(\mathfrak{M})\}.$$

This group clearly depends on the choice of the coloring χ. However, if χ' is another $\mathfrak{M}$-coloring of $\mathbf{T}$, the groups obtained using χ and χ' are conjugate in $\mathrm{Aut}(\mathbf{T})$ by any element of $\mathrm{Aut}(\mathbf{T})$ changing χ into χ', which always exists.

This construction is the infinite-arity analogue of the one carried out in [BM00, Sec. 3]. The terminology is motivated by the following (adapted from the original result [BM00, Prop. 3.2.2.] in the finite arity case):

Proposition 1.2. *Suppose* $\mathfrak{M}$ *is transitive and let* $H \leqslant \mathrm{Aut}(\mathbf{T})$ *be a vertex-transitive subgroup such that for every* $x \in \mathbf{T}$, *the action* $H \curvearrowright E_{\mathbf{T}}(x)$ *is isomorphic to* $\Gamma \curvearrowright \mathfrak{M}$. *Then* H *is contained in a conjugate of* $\mathrm{Aut}_{\mathfrak{M}}(\mathbf{T})$ *by an element of* $\mathrm{Aut}(\mathbf{T})$.

Interestingly, this construction predates [BM00] by at least two decades. Indeed, it appears in [Ol'80] where Ol'šanskiĭ classifies the irreducible continuous unitary representations of these groups in the finite-arity case assuming 2-transitivity. The purpose of the present paper is to get a similar classification for infinite-arity Burger-Mozes universal groups.

Convention. From now on, we fix a countably-infinite structure $\mathfrak{M}$, an $\mathfrak{M}$-coloring χ and still denote by $\mathbf{T}$ the resulting colored tree. χ is suppressed from all notations. The colored tree $\mathbf{T}$ is represented in Figure 1a where $x,y,z \in \mathbf{T}$, $a,b \in \mathfrak{M}$ and $\chi(x,y) = a$, $\chi(x,z) = b$. **In the remaining of this paper, we will write** Γ **instead of** $\mathrm{Aut}(\mathfrak{M})$ **and** G **instead of** $\mathrm{Aut}_{\mathfrak{M}}(\mathbf{T})$**.** Moreover, we will call $\Gamma \curvearrowright \mathfrak{M}$ the *local action* and $G \curvearrowright \mathbf{T}$ the *global action*. Any additional hypothesis on $\mathfrak{M}$ will be specified in each result separately.

We start the study of these groups with a general method for building elements of G. All one needs to provide is the image of an arbitrary vertex and a coherent collection of local actions:

Lemma 1.3. *Let* $x,y \in \mathbf{T}$ *and let* $(\gamma_u)_{u \in \mathbf{T}} \in \Gamma^{\mathbf{T}}$. *The following assertions are equivalent:*

(i) there exists a unique $g \in G$ *such that* $g(x) = y$ *and* $g_u = \gamma_u$ *for all* $u \in \mathbf{T}$,
(ii) for every $u \sim v$ *in* $\mathbf{T}$, $\gamma_u(\chi(u,v)) = \gamma_v(\chi(u,v))$.

Proof. The implication $(i) \Rightarrow (ii)$ is trivial. For the converse, let $(\gamma_u)_{u\in\mathbf{T}}$ be as in (ii). We define g inductively.

- Set $g(x) = y$.
- Suppose g is defined on $B(x,n)$ for some $n \in \mathbb{N}$ in such a way that:
  1. $g\colon B(x,n) \longrightarrow B(y,n)$ is a graph isomorphism,
  2. $\forall u \in B(x,n)\backslash S(x,n),\ g_u = \gamma_u$.

  Given $u \in \mathbf{T}$ such that $d(u,x) = n+1$, there exists a unique $v \in B(x,n)$ which is connected to u. Define $g(u)$ as the unique $u' \in V_{\mathbf{T}}(g(v))$ such that $\chi(u',g(v)) = \gamma_v(\chi(u,v))$. Once this is done for every u at distance $n+1$ from x, g is defined on $B(x,n+1)$. Because $(\gamma_z)_{z\in\mathbf{T}}$ satisfies (ii), Items 1. and 2. above still hold after replacing n with $n+1$.

This proves existence. Uniqueness is straightforward.

□

Corollary 1.4. *Let $x,y \in \mathbf{T}$ and let X be a subtree of $\mathbf{T}$. Let $(\gamma_z)_{z\in X} \in \Gamma^X$ be such that for every $u \sim v$ in X, $\gamma_u(\chi(u,v)) = \gamma_v(\chi(u,v))$.*

There exists $g \in G$ such that $g(x) = y$ and $g_u = \gamma_u$ for all $u \in X$.

Proof. If X is empty, apply Lemma 1.3 to $(\mathrm{id}_{\mathbf{T}})_{u\in\mathbf{T}}$. If not, we will extend $(\gamma_u)_{u\in X}$ to $\mathbf{T}$ by propagation: Given $u \in \mathbf{T}\backslash X$, let $p(u)$ be the unique element of X such that $d(u,p(u)) = d(u,X)$ and set $\gamma_u = \gamma_{p(u)}$. It now suffices to check that $(\gamma_u)_{u\in\mathbf{T}}$ satisfies condition (ii) of Lemma 1.3. To that aim, let $u \sim v$ be adjacent vertices in $\mathbf{T}$. There are three case:

- If both u and v are in X, then $\gamma_u(\chi(u,v)) = \gamma_v(\chi(u,v))$ by assumption.
- If neither are in X, then $p(u) = p(v)$ since $u \sim v$, hence $\gamma_u = \gamma_v$.
- Otherwise, we can assume that $u \notin X$ and $v \in X$. Then $p(u) = v$ and $\gamma_u = \gamma_v$.

□

Here are the different notions of transitivity we will be working with.

Definition 1.5. $\mathfrak{M}$ is said to be

1. *transitive* if the action $\Gamma \curvearrowright \mathfrak{M}$ has only one orbit.
2. *2-transitive* if for every $a \neq b$, $a' \neq b' \in \mathfrak{M}$, there exists $\gamma \in \Gamma$ such that $\gamma(a) = a'$ and $\gamma(b) = b'$.

Similarly, given $H \leqslant \mathrm{Aut}(\mathbf{T})$, the action of H on $\mathbf{T}$ is said to be:

1. *vertex transitive* if it has only one orbit.
2. *edge transitive* if for every $x,y,x',y' \in \mathbf{T}$ with $x \sim y$ and $x' \sim y'$, there exists $h \in H$ such that $h(x) = x'$ and $h(y) = y'$.
3. *2-transitive* if for every $x,y,x',y' \in \mathbf{T}$ with $d(x,y) = d(x',y')$, there exists $h \in H$ such that $h(x) = x'$ and $h(y) = y'$.

We will be going back and forth between the global action $G \curvearrowright \mathbf{T}$ and the local action $\Gamma \curvearrowright \mathfrak{M}$. Given $x \in \mathbf{T}$, a set $A \subseteq E_{\mathbf{T}}(x)$ of edges around x identifies through the coloring χ with a subset of $\mathfrak{M}$. Next is how we will make identifications in the other direction:

Definition 1.6. Given $a \in \mathfrak{M}$ and $x \in \mathbf{T}$, we will denote by a_x the unique vertex of $\mathbf{T}$ adjacent to x such that $\chi(x,a_x) = a$. Similarly, if A is a subset of $\mathfrak{M}$, it identifies at x with the following star-shaped subtree of $\mathbf{T}$:

$$A_x := \{x\} \cup \{y \in V_{\mathbf{T}}(x)\colon \chi(x,y) \in A\}\,.$$

As expected, there is a clear correspondence between local and global transitivity properties.

Lemma 1.7. *The following properties hold:*

1. *$G \curvearrowright \mathbf{T}$ is vertex transitive.*
2. *$G \curvearrowright \mathbf{T}$ is edge transitive $\iff$ $\Gamma \curvearrowright \mathfrak{M}$ is transitive.*
3. *$G \curvearrowright \mathbf{T}$ is 2-transitive $\iff$ $\Gamma \curvearrowright \mathfrak{M}$ is 2-transitive.*

Proof.

1. Let $x,y \in \mathbf{T}$. Simply apply Corollary 1.4 to $X = \{x\}$ and $\gamma_x = \mathrm{id}_{\mathbf{T}}$ while requiring x to be sent to y.

2. Suppose $\mathfrak{M}$ is transitive. Let $x \sim y, x' \sim y' \in \mathbf{T}$. By hypothesis, there exists $\gamma \in \Gamma$ such that $\gamma(\chi(x,y)) = \chi(x',y')$. Set $X := \{x,y\}$ and $\gamma_x = \gamma_y = \gamma$. Apply Corollary 1.4 while requiring x to be sent to y. The converse is straightforward.
3. Suppose $G \curvearrowright \mathbf{T}$ is 2-transitive and let $a \neq b, a' \neq b' \in \mathfrak{M}$. Fix any $x \in \mathbf{T}$. Then:
$$d(a_x, b_x) = 2 = d(a'_x, b'_x)\ ,$$
hence there exists $g \in G$ such that $g(a_x) = a'_x$ and $g(b_x) = b'_x$. Necessarily, $g(x) = x$ and g_x is an element of Γ that sends a to a' and b to b'.

   Conversely, assume $\Gamma \curvearrowright \mathfrak{M}$ is 2-transitive. Let $x, y, x', y' \in \mathbf{T}$ be vertices satisfying $d(x,y) = d(x',y')$. Write $[x,y] = (x_0, \dots, x_n)$ and $[x',y'] = (x'_0, \dots, x'_n)$. For every $i \in \{1, \dots, n\}$, let $a_i := \chi(x_{i-1}, x_i)$ and $a'_i := \chi(x'_{i-1}, x'_i)$. By hypothesis on $\mathfrak{M}$, there exists for every $i \in \{1, \dots, n-1\}$ some element $\gamma_i \in \Gamma$ such that $\gamma_i(a_i) = a'_i$ and $\gamma_i(a_{i+1}) = a'_{i+1}$.

   We now apply Corollary 1.4 to $X = \{x_1, \dots, x_{n-1}\}$ and $\gamma_{x_i} = \gamma_i$ for every $i \in \{1, \dots, n-1\}$ while requiring x_1 to be sent to x'_1. Visibly, the output g sends both x to x' and y to y'.

□

Unfortunately, the action $G \curvearrowright \mathbf{T}$ is not rich enough to fully capture the representation theory of G in general. In fact, the lack resides in the local action $\Gamma \curvearrowright \mathfrak{M}$. In order to fix this, we have to take the imaginaries of $\mathfrak{M}$ into account. Recall that an *imaginary element* on $\mathfrak{M}$ is the equivalence class of a tuple with respect to some first-order definable equivalence relation over $\mathfrak{M}^n$. For every formula $\phi(\overline{x}, \overline{y})$ defining an equivalence relation on $\mathfrak{M}^n$, add to the language of $\mathfrak{M}$ both a sort S_ϕ and a function symbol p_ϕ of arity n. Then $\mathfrak{M}^{\mathrm{eq}}$ is the natural expansion of $\mathfrak{M}$ to this language, where $\mathfrak{M}^n/\phi$ is the sort corresponding to S_ϕ and where the interpretation of p_ϕ is the map $\mathfrak{M}^n \longrightarrow \mathfrak{M}^n/\phi$ that sends an n-tuple to its equivalence class. It is a classical fact that the action $\mathrm{Aut}(\mathfrak{M}) \curvearrowright \mathfrak{M}$ naturally extends to an action on $\mathfrak{M}^{\mathrm{eq}}$, yielding a canonical identification $\mathrm{Aut}(\mathfrak{M}) \simeq \mathrm{Aut}(\mathfrak{M}^{\mathrm{eq}})$ (see [Hod93, Th. 4.3.3]). Note that since the language is countable, one can only write countably many formulae hence $\mathfrak{M}^{\mathrm{eq}}$ remains countable. Note also that $\mathfrak{M}^{\mathrm{eq}} \backslash \mathfrak{M}$ is infinite as it contains a copy of $\mathfrak{M}^n$ for every $n > 1$.

Imaginaries are closely related to representation theory. Indeed, the elements of $\mathfrak{M}^{\mathrm{eq}}$ are in correspondence with the open subgroups of Γ via the map $e \longmapsto \Gamma_e$. In particular, the action $\Gamma \curvearrowright \mathfrak{M}^{\mathrm{eq}}$ is richer than the original action $\Gamma \curvearrowright \mathfrak{M}$ in the sense that it captures more open subgroups, if not all. This is essential as every open subgroup U of Γ gives rise to a unitary representation of Γ. Indeed, Γ/U is countable and we have the action $\Gamma \curvearrowright \ell^2(\Gamma/U)$.

We now implement this expansion of $\mathfrak{M}$ into our construction. To that aim, let $\mathbf{T}^{\mathrm{eq}}$ be the tree obtained from $\mathbf{T}$ by adding a countably infinite collection of leaves around each of its vertices. The coloring χ extends to an $\mathfrak{M}^{\mathrm{eq}}$–coloring of $\mathbf{T}^{\mathrm{eq}}$ in such a way that:
$$\forall x \in \mathbf{T},\ [\chi \colon E_{\mathbf{T}^{\mathrm{eq}}}(x) \longrightarrow \mathfrak{M}^{\mathrm{eq}} \text{ is a bijection}].$$
The resulting colored tree will still be denoted $\mathbf{T}^{\mathrm{eq}}$ and is represented in Figure 1b. We extend Definition 1.6 to that context. The previously mentioned identification $\mathrm{Aut}(\mathfrak{M}) \simeq \mathrm{Aut}(\mathfrak{M}^{\mathrm{eq}})$ implies that $G \curvearrowright \mathbf{T}$ naturally extends to a richer action $G \curvearrowright \mathbf{T}^{\mathrm{eq}}$.

Convention. In graphical representations such as in Figure 1, solid vertices and thick edges will always correspond to elements of $\mathbf{T}$ or labels in $\mathfrak{M}$. On the other hand, hollowed vertices and thin edges will always correspond to elements of $\mathbf{T}^{\mathrm{eq}} \backslash \mathbf{T}$ or labels in $\mathfrak{M}^{\mathrm{eq}} \backslash \mathfrak{M}$.

Remark 1.8. Enlarging $\mathbf{T}$ into $\mathbf{T}^{\mathrm{eq}}$ is not required if $\mathfrak{M}$ admits *weak elimination of imaginaries* (abbreviated w.e.i.). We refer to [Hod93, Sec. 4.4] for the formal definition and simply recall that if $\mathfrak{M}$ is $\aleph_0$-categorical, it admits w.e.i. if and only if for every open subgroup U of Γ there exists a finite subset $A \subseteq \mathfrak{M}$ such that:
$$G_A \leqslant U \leqslant G_{\{A\}}. \tag{1}$$
In other words, an $\aleph_0$-categorical structure $\mathfrak{M}$ admits w.e.i. if and only if the action $\Gamma \curvearrowright \mathfrak{M}$ already captures all of the open subgroups of Γ up to finite index. Only assuming $\mathfrak{M}$ to be $\aleph_0$-categorical, $\mathfrak{M}^{\mathrm{eq}}$ has the stronger property that the map $e \longmapsto \Gamma_e$ is surjective onto the open subgroups of Γ. In fact, there is a purely group theoretic interpretation of $\mathfrak{M}^{\mathrm{eq}}$ in this context.

Indeed, if Γ is oligomorphic, then it only has countably many distinct open subgroups [Tsa12, Cor. 2.5]. Moreover, the action $\Gamma \curvearrowright \mathfrak{M}^{\mathrm{eq}}$ is essentially the action of Γ on the countable set of its open cosets.

This step is also redundant for groups that are dissociated in the sense of [BJJ24]. Indeed, dissociated groups satisfy (1). As soon as (1) holds, the statements and proofs in Section 2 can all be simplified and worked out directly in $\mathbf{T}$.

Lemma 1.9. *Let X be a non-empty subtree of $\mathbf{T}^{\mathrm{eq}}$ and let $x \in X$. Let $Z := X \cup V_{\mathbf{T}^{\mathrm{eq}}}(x)$ and let $A := E_X(x)$, seen as a subset of $\mathfrak{M}^{\mathrm{eq}}$. The restriction morphism*

$$r_x \colon G_X \longrightarrow \Gamma_A\,, \qquad g \longmapsto g_x$$

is continuous, open and surjective. In particular, $G_X/G_Z \simeq \Gamma_A$ (as topological groups).

Proof. Continuity is clear. Regarding openness, it suffices to prove that images of small open subgroups are open. To that aim, let Y be a finite subtree of $\mathbf{T}^{\mathrm{eq}}$ containing X and let $Y' := Y \cap \mathbf{T}$. Let $B := E_Y(x)$, seen as a subset of $\mathfrak{M}^{\mathrm{eq}}$. Given $\gamma \in \Gamma_B$, set $\gamma_x := \gamma$ and $\gamma_y := \mathrm{id}$ for every $y \in Y' \backslash \{x\}$. Then $(\gamma_y)_{y \in Y'}$ satisfies the conditions of Corollary 1.4. We thus get $g \in G$ such that $g(x) = x$ and for every $y \in Y'$, $g_y = \gamma_y$. In particular, $g_x = \gamma$ and, visibly, $g \in G_Y$. Thus $r_x(G_Y) = \Gamma_B$, showing openness. Specifying to $Y = X$, we also get surjectivity. The kernel of r_x being precisely G_Z, the last claim follows. □

Corollary 1.10. *Let X be a subtree of $\mathbf{T}^{\mathrm{eq}}$ and $x \in X$. Let $A := E_X(x)$ be seen as a subset of $\mathfrak{M}^{\mathrm{eq}}$. For every $y \in V_{\mathbf{T}^{\mathrm{eq}}}(x)$, the map*

$$G_X \cdot y \longrightarrow \Gamma_A \cdot \chi(x,y)\,, \qquad g(y) \longmapsto g_x(\chi(x,y))$$

is a bijection.

As in the finite-arity case (see [BM00, Sec. 3.2] for the proof), one can give a precise structural description of vertex stabilizers in G.

Proposition 1.11. *Assume $\mathfrak{M}$ is transitive and fix $a \in \mathfrak{M}$ and $x \in \mathbf{T}$. For every $n \geqslant 1$, let $G_n \leqslant \mathrm{Sym}(S_{\mathbf{T}}(x,n))$ be the permutation group induced by the action of G_x on $S_{\mathbf{T}}(x,n)$. Then:*

1. *$(G_n)_{n \geqslant 1}$ is naturally an inverse sequence of which G is the inverse limit,*
2. *$G_1 \simeq \Gamma$ and for every $n \geqslant 1$, $G_{n+1} \simeq G_n \ltimes \Gamma_a^{S(x,n)}$ (as permutation groups).*

1.2. $\aleph_0$-categoricity, Roelcke precompactness, and coarse geometry.

The representation theory of G is at least as complicated as that of Γ. Indeed, if $\sigma \colon \Gamma \curvearrowright \mathcal{H}$ is a unitary representation of Γ, then it can be pulled back to a representation of G_x via the restriction morphism $G_x \longrightarrow \Gamma$ given by $g \longmapsto g_x$. Since this morphism is surjective by Lemma 1.9, this representation of G_x retains all the richness of σ and can in turn be induced to G. For this reason, we must restrict our attention to instances of the construction where the representation theory of Γ is understood. Recall that $\mathfrak{M}$ is $\aleph_0$-categorical if and only $\Gamma \curvearrowright \Omega$ is oligomorphic and that the representations of oligomorphic groups have been fully classified in [Tsa12]. We will restrict our attention to this case.

The topological manifestation of $\aleph_0$-categoricity is Roelcke precompactness. Recall that a subset E of a topological group H is *Roelcke precompact* if for every open neighborhood U of the identity in H there exists a finite subset $F \subseteq H$ such that $E \subseteq UFU$. A group is *Roelcke precompact* if it is Roelcke precompact as a subset of itself and it is *locally Roelcke-precompact* if it admits a Roelcke-precompact non-empty open subset. It is an easy consequence of the Ryll-Nardzewski Theorem (e.g. in combination with [BYT16, Th. 2.4]) that for $\mathfrak{M}$ transitive, $\mathrm{Aut}(\mathfrak{M})$ is Roelcke precompact exactly when $\mathfrak{M}$ is $\aleph_0$-categorical. In other words, if $\Gamma \curvearrowright \Omega$ is transitive, then Γ is oligomorphic if and only if it is Roelcke precompact.

Roelcke precompactness is a generalization of compactness that seems to retain most of its geometrical properties. Within representation theory, this is witnessed by Tsankov's classification Theorem [Tsa12] which is reminiscent of the Peter-Weyl Theorem for compact groups. Another strong similarity is that every Roelcke precompact Polish group has Kazdhan's Property

(T) [Iba21]. The duality theory developed by Tannaka and Krein for compact groups using harmonic analysis also extends to Roelcke precompact automorphism groups [Bar25]. Recall that $G^+ = \{g \in G\colon \forall x \in \mathbf{T},\ d(g(x), x) \text{ is even}\}$

Proposition 1.12. *Assume $\mathfrak{M}$ is transitive and $\aleph_0$-categorical. Then G and G^+ are locally Roelcke-precompact. More precisely, the stabilizers of vertices and edges in G are Roelcke precompact.*

Proof. For vertex stabilizers, this is a consequence of Proposition 1.11. Indeed, direct products, semi-direct products and inverse limits of Roelcke precompact groups are Roelcke precompact [Tsa12, Prop. 2.2].

Let $x, y \in \mathbf{T}$ be connected. Since $\mathfrak{M}$ is transitive, $G \curvearrowright \mathbf{T}$ is edge transitive by Item 2 of Lemma 1.7 and there exists $g \in G$ that exchanges x and y. Then $G_{\{\{x,y\}\}} = G_{\{x,y\}} \sqcup gG_{\{x,y\}} \leqslant G_x \cup gG_x$ hence $G_{\{\{x,y\}\}}$ is Roelcke precompact as a subset of a finite union of Roelcke precompact sets. □

The above proposition highlights another benefit of inputting $\aleph_0$-categorical structures: it produces particularly well-behaved groups in terms of Rosendal's *large scale geometry* [Ros22]. Recall that an *écart* on a topological group H is a map $\delta\colon H^2 \longrightarrow \mathbb{R}_+$ such that for every $x, y, z \in H$, we have $\delta(x,x) = 0$ and $\delta(x,z) \leqslant \delta(x,y) + \delta(y,z)$. It is *left-invariant* if $\delta(xy, xz) = \delta(y,z)$ for every $x, y, z \in H$. A subset $E \subseteq H$ is *coarsely bounded* (in the left coarse-structure) if its δ-diameter $\delta(E)$ is finite for every continuous left-invariant écart δ on H.

It is easily seen that Roelcke precompact subsets are always coarsely bounded. In particular, Roelcke precompact groups have trivial large scale geometry. It turns out, however, that algebra and topology blend most harmoniously in the large scale geometry of locally Roelcke-precompact Polish groups [Ros22, Chap. 3]. For one thing, these groups are characterized by having a locally compact Roelcke completion [Zie21], see also Subsection 3.1 below. What is more, a subset of a locally Roelcke-precompact Polish group H is coarsely bounded if and only if it is Roelcke precompact. Finally, such a group H always admits a *coarsely proper* écart, i.e. a left-invariant and continuous écart δ that generates the left coarse-structure in the following sense:

$$\forall E \subseteq H,\ [E \text{ is coarsely bounded} \iff \delta(E) < \infty].$$

In our case, the écart naturally arising from the action $G \curvearrowright \mathbf{T}$ plays such a role.

Corollary 1.13. *Assume $\mathfrak{M}$ is 2-transitive and $\aleph_0$-categorical. Given K be a subset of G (resp. G^+), the following properties are equivalent:*

(i) K is coarsely bounded in G (resp. G^+),
(ii) K is Roelcke precompact in G (resp. G^+),
(iii) $K \cdot x$ is bounded in $\mathbf{T}$ for every vertex $x \in \mathbf{T}$,
(iv) $K \cdot x$ is bounded in $\mathbf{T}$ for some vertex $x \in \mathbf{T}$,

If K is a subgroup of G (or G^+), the previous properties are moreover equivalent to:

(v) $K \leqslant G_x$ for some vertex $x \in \mathbf{T}$ or $K \leqslant G_{\{\{x,y\}\}}$ for some connected vertices $x, y \in \mathbf{T}$,
(vi) $K \leqslant G_{\{X\}}$ for some non-empty finite subtree $X \subseteq \mathbf{T}$.

Proof. The proofs for G and G^+ are the same since $G_A = G_A^+$ as soon as $A \subseteq \mathbf{T}$ is non-empty, hence we only prove the equivalences for G.

$\underline{(i) \iff (ii)}$: Since G is locally Roelcke-precompact by Proposition 1.12, we refer to [Zie21, Th. 14] for this equivalence.

$\underline{(ii) \Rightarrow (iii)}$: Assume K is Roelcke precompact in G and let $x \in \mathbf{T}$. Then, G_x being open, there exists a finite subset F of K such that $K \subseteq G_x F G_x$. Let $M := \max\{d(f(x), x)\colon f \in F\}$. If $k \in K$, there exists $u, v \in G_x$ and $f \in F$ such that $k = ufv$. Then $d(k(x), x) = d(fv(x), u^{-1}(x)) = d(f(x), x) \leqslant M$.

$\underline{(iii) \Rightarrow (iv)}$: Trivial.

$\underline{(iv) \Rightarrow (ii)}$: Let $x \in \mathbf{T}$ be such that $K \cdot x$ is bounded. We first observe that, for every $g \in G$, the double coset $G_x g G_x$ is determined by $d(gx, x)$. Indeed, suppose $g, h \in G$ satisfy $d(gx, x) = d(hx, x)$. Then, since $G \curvearrowright \mathbf{T}$ is 2-transitive by Item 3 of Lemma 1.7, there exists $u \in G$ such that $ug(x) = h(x)$ and $u(x) = x$. In particular, $h^{-1}ug \in G_x$ and $g \in G_x h G_x$ i.e. $G_x g G_x = G_x h G_x$.

Now, since $K \cdot x$ is bounded, the set $\{d(kx, x) \colon k \in K\}$ is finite. Fix $F \subseteq K$ finite covering all the possibilities. By what precedes, $K \subseteq G_x F G_x$. Now, G is locally Roelcke-precompact by Proposition 1.12 hence products of Roelcke precompact subsets of G are Roelcke precompact [Zie21, Prop. 12]. Since G_x is Roelcke precompact in G by Proposition 1.12 and since finite sets are always Roelcke precompact, so is K.

We now assume that K is a subgroup of G. The implications $(v) \Rightarrow (vi) \Rightarrow (iii)$ being obvious if even K is not a subgroup, it only remains to prove $(iv) \Rightarrow (v)$. Suppose $x \in \mathbf{T}$ is such that $K \cdot x$ is bounded. Let $X := \langle K \cdot x \rangle$ be the tree generated by $K \cdot x$. This tree has finite diameter and is stable under the action of the group K. If X has diameter 0 or 1, then (v) holds. If not, let X' be the tree obtained from X by removing all of its leaves. Then X' remains stable under the action of K while the diameter has decreased by 2. We conclude by induction on the diameter of X. □

In Section 3, we will see that the large scale geometry of G is well captured by its representation theory, provided that $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. More precisely, G^+ has the Howe-Moore property.

1.3. **Algebraicity.**

The notion of algebraicity will be central in our study. We will be working with its permutation group theoretic definition. Fortunately, when $\mathfrak{M}$ is $\aleph_0$-categorical, it coincides with the model theoretic definition, both for $\Gamma \curvearrowright \mathfrak{M}$ and $\Gamma \curvearrowright \mathfrak{M}^{\mathrm{eq}}$.

Definition 1.14. Let H be a closed permutation group over a set Ω and let A be a subset of Ω.

1. An element $x \in \Omega$ is said to be *algebraic over A* if there exists a finite subset $B \subseteq A$ such that the orbit $H_B \cdot x$ is finite. The *algebraic closure* of A, denoted $\mathrm{acl}_\Omega(A)$, is the set of elements of Ω that are algebraic over A. The set A is *algebraically closed* if it is equal to its algebraic closure. Recall that algebraic closures are always algebraically closed.
2. A is said to be *definable* if $H_{\{D\}}$ is open in H.
3. A is said to be *locally finite* if its intersection with every orbit of the action of H on Ω is finite.

We will write $\mathcal{C}(\Omega)$ for the set of definable, locally finite and algebraically closed subsets of Ω, called the *classifying subsets*. When $\mathfrak{M}$ is $\aleph_0$-categorical, then $(\mathcal{C}(\mathfrak{M}^{\mathrm{eq}}), \subseteq, \wedge, \vee)$ where $A \wedge B = A \cap B$ and $A \vee B = \mathrm{acl}_\Omega(A \cup B)$ for all $A, B \in \mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ is an ordered lattice [Tsa25, Th. 2.4]. In this case, understanding $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ is key to understanding the representation theory of Γ, see the end of the next subsection. Below are the properties of this lattice that will be useful to us, extracted from [Tsa25].

Proposition 1.15 ([Tsa25])**.** *Assuming $\mathfrak{M}$ is $\aleph_0$-categorical, the following properties hold:*

1. *The elements of $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ are exactly the subsets of $\mathfrak{M}^{\mathrm{eq}}$ of the form $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(e)$ for some $e \in \mathfrak{M}^{\mathrm{eq}}$ (equivalently, of the form $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)$ for some finite subset A of $\mathfrak{M}^{\mathrm{eq}}$).*
2. *For every $A \in \mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$, $A \cap \mathfrak{M}$ is finite.*
3. *The ordered lattice $(\mathcal{C}(\mathfrak{M}^{\mathrm{eq}}), \subseteq, \wedge, \vee)$ is well-founded, i.e. every non-empty subset of $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ admits a minimal element.*

Proof.

1. This is the combination of Lemmas 2.1 and 2.2 in [Tsa25].
2. Sin $\mathfrak{M}$ is $\aleph_0$-categorical, $\gamma \curvearrowright \mathfrak{M}$ is oligomorphic and $\mathfrak{M}$ splits into finitely many Γ-orbits. This item then follows from the fact that elements of $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ are locally finite.
3. This is [Tsa25, Th. 2.4].

□

The smallest element of $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ is obviously $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. Assuming $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive, every element of

$$\mathcal{C}_1(\mathfrak{M}^{\mathrm{eq}}) := \{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a),\ a \in \mathfrak{M}\}$$

is a direct successor of $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. We end this subsection with the proof of this fact.

Lemma 1.16. *Let Ω be an infinite countable set and let $(E_i)_{i\in I}$ be a countably infinite collection of pairwise distinct finite subsets of Ω. Suppose there exists $n \in \mathbb{N}$ such that $|E_i \cap E_j| = n$ for every $i \neq j \in I$. Then:*

$$\forall i \neq j \in I,\ E_i \cap E_j = \bigcap_{k\in I} E_k.$$

Proof. Let $i_1 \in I$. The set $\{E_{i_1} \cap E_j,\ j \in I\}$ is an infinite collection of subsets of the finite set E_{i_1}. By the pigeonhole principle, there exists an infinite subset $J \subseteq I$ such that for every $j, j' \in J$, $E_{i_1} \cap E_j = E_{i_1} \cap E_{j'}$. We will denote by F this common intersection.

Now, for every $j \neq j' \in J$,

$$F = F \cap F = E_{i_1} \cap E_j \cap E_{i_1} \cap E_{j'} \subseteq E_j \cap E_{j'}.$$

Since moreover $|F| = n = |E_j \cap E_{j'}|$, we have:

$$\forall j \neq j' \in J,\ E_j \cap E_{j'} = F.$$

Let $i_2 \in I\backslash\{i_1\}$. The set $\{E_{i_2} \cap E_j,\ j \in J\}$ is an infinite collection of subsets of the finite set E_{i_2}. By the pigeonhole principle, there exists an infinite subset $K \subseteq J$ such that for every $k, k' \in K$, $E_{i_2} \cap E_k = E_{i_2} \cap E_{k'}$. It follows from the argument in the previous paragraph that this common intersection is precisely F.

Finally, K being infinite, we can pick $k \in K\backslash\{i_1, i_2\}$. Recalling that $K \subseteq J$, we have:

$$F = F \cap F = E_{i_1} \cap E_k \cap E_{i_2} \cap E_k \subseteq E_{i_1} \cap E_{i_2}.$$

By cardinality, $E_{i_1} \cap E_{i_2} = F$. Since i_1 and i_2 are arbitrary, the conclusion follows. □

Corollary 1.17. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $a \in \mathfrak{M}$ and $A \in \mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ be such that $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset) \subseteq A \subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$. Then either $A = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$ or $A = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$.*

Proof. Let $n \in \mathbb{N}$ and let $\varphi(\overline{x}, \overline{y})$ be a formula of arity $2n$ encoding an equivalence relation on $\mathfrak{M}^n$. Let $\Omega := \mathfrak{M}^n/\varphi$ be the associated set of equivalence classes in $\mathfrak{M}^n$. By the Ryll-Nardzewski Theorem, the action $\Gamma \curvearrowright \mathfrak{M}$ has finitely many orbits hence so does $\Gamma \curvearrowright \Omega$. Since Γ is Roelcke precompact, the action $\Gamma_a \curvearrowright \Omega$ also has finitely many orbits for every $a \in \mathfrak{M}$. In particular, $E_a := \{x \in \Omega : \Gamma_a \cdot x \text{ is finite}\}$ is finite for every $a \in \mathfrak{M}$. What is more, by 2-transitivity of $\mathfrak{M}$, $|E_a \cap E_b|$ does not depend on the choice of $a \neq b \in \mathfrak{M}$. By Lemma 1.16, neither does $E_a \cap E_b$. Since φ was arbitrary, we have in fact proved that for all $a \neq b \in \mathfrak{M}$, $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a) \cap \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(b) = C := \bigcap_{c\in\mathfrak{M}} \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(c)$.

We claim that $C = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. Indeed, let $a \neq b \in \mathfrak{M}$. By 2-transitivity of $\mathfrak{M}$, $\Gamma_a\Gamma_b \cdot a = \Gamma_a \cdot (\mathfrak{M}\backslash\{b\}) = \mathfrak{M}$ thus $\Gamma = \Gamma_a\Gamma_b\Gamma_a$. Suppose now that $c \in C$. Since C is stable under the action of Γ, $F_1 := \Gamma_a \cdot c$ is a finite subset of C hence so is $F_2 := \bigcup \Gamma_b \cdot F_1$ and so is $F_3 := \bigcup \Gamma_a \cdot F_2$. Since $F_3 = \Gamma_a\Gamma_b\Gamma_a \cdot c = \Gamma \cdot c$, we indeed get that $c \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. The reverse inclusion is trivial. We have thus shown that

$$\forall a \neq b \in \mathfrak{M},\ \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a) \cap \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(b) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset).$$

We are now ready to conclude. Let $e \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$. It suffices to show that either $a \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(e)$ or $e \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. Suppose $a \notin \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(e)$. In particular, there exists $\gamma \in \Gamma_e$ such that $b := \gamma(a) \neq a$. The equality $e = \gamma(e)$ yields $e \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a) \cap \gamma(\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a) \cap \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(b) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$.

□

1.4. **Hilbert spaces and unitary representations.**

We recall here some basic results about Hilbert spaces and unitary representations of permutation groups. First, the celebrated Alaoglu-Birkhoff Theorem will help us locate invariant vectors.

Theorem 1.18. *Let $\mathcal{H}$ be a Hilbert and let K be any subgroup of $\mathcal{U}(\mathcal{H})$. Let p be the orthogonal projection on $\mathcal{H}^K$, the closed subspace of K-invariant vector. Then, for every $\eta \in \mathcal{H}$, $p\eta$ belongs to the norm-closed convex hull $\overline{\mathrm{Conv}}(K \cdot \eta)$ of $K \cdot \eta$. More precisely, $p\eta$ is the only K-invariant vector in $\overline{\mathrm{Conv}}(K \cdot \eta)$.*

Next is a convergence theorem for projections associated with a directed collection of subspaces of a Hilbert space, which will allow us to carry out approximation arguments.

Lemma 1.19. *Let $\mathcal{H}$ be a Hilbert space, $(I,\leqslant)$ a directed set and $(\mathcal{K}_i)_{i\in I}$ a directed union of closed subspaces of $\mathcal{H}$. For every $i\in I$, denote by p_i the orthogonal projection onto $\mathcal{K}_i$. Let $\mathcal{K}:=\overline{\bigcup_{i\in I}\mathcal{K}_i}$ and let p be the orthogonal projection onto $\mathcal{K}$. Then $(p_i)_{i\in I}$ converges to p in the strong operator topology, that is:*

$$\forall \xi\in\mathcal{H},\ \|p_i\xi-p\xi\|\underset{i}{\longrightarrow}0\ .$$

We now recall classical definitions and facts about unitary representations. A *unitary representation* of a topological group K is a linear isometric action of K on a Hilbert space $\mathcal{H}$ that is. It is continuous if for every $\xi,\eta\in\mathcal{H}$, the map $k\longmapsto\langle\pi(k)\xi,\eta\rangle$ is continuous. It is *non-zero* if $\mathcal{H}\neq 0$. Throughout this paper, "representation" will always mean "continuous unitary representation". Let $\pi\colon K\curvearrowright\mathcal{H}$ and $\pi'\colon K\curvearrowright\mathcal{H}'$ be two representations of K. An *isomorphism* between π and π' is a K-equivariant surjective isometry $\mathcal{H}\longrightarrow\mathcal{H}'$. We will say that π' is (isomorphic to) a *subrepresentation* of π, and write $\pi'\subseteq\pi$, if there exist a K-equivariant isometric embedding $\mathcal{H}'\longrightarrow\mathcal{H}$. The subrepresentation π' is *proper* if $\mathcal{H}'$ identifies with a proper subspace of $\mathcal{H}$ via the embedding. The representation π is *irreducible* if $\mathcal{H}\neq 0$ and π admits no non-zero proper subrepresentation. The *unitary dual* of K is the set of isomorphism classes of irreducible representations of K.

Induction is a central technique in harmonic analysis. Suppose U is an open subgroup of a separable topological group K, then K/U is countable discrete and can be endowed with the counting measure. If moreover $\sigma\colon U\curvearrowright\mathcal{H}$ is a unitary representation of U, let $E:=\{\xi\colon K\longrightarrow\mathcal{H}\mid\forall u\in U,\forall k\in K,\ \xi(ku)=\sigma(u^{-1})\cdot\xi(k)\}$. Then, for every $\xi,\eta\in E$, the maps defined on K by

$$k\longmapsto\langle\xi(k),\eta(k)\rangle\qquad\&\qquad k\longmapsto\|\xi(k)\|$$

are constant on every U-coset. Fix a transversal $(k_i)_{i\in I}$ of K/U and define:

$$\mathcal{E}:=\{\xi\in E\colon\ \sum_{i\in I}\|\xi(k_i)\|<+\infty\}.$$

Then $\mathcal{E}$ is a Hilbert space with the following inner product:

$$\forall\xi,\eta\in\mathcal{E},\ \langle\xi,\eta\rangle:=\sum_{i\in I}\langle\xi(k_i),\eta(k_i)\rangle.$$

In this setting, we will write $\xi\in\ell^2(G/U)$ to refer to the fact that if $\xi\in\mathcal{E}$, then $\sum_{i\in I}\|\xi(k_i)\|<+\infty$. The *induced representation* $\mathrm{Ind}_U^K(\sigma)$ is the action $K\curvearrowright\mathcal{E}$ defined by

$$\forall k\in K,\forall\xi\in\mathcal{E},\quad k\cdot\xi\colon x\longmapsto\xi(k^{-1}x).$$

Note that this construction does not depend on the choice of a transversal $(k_i)_{i\in I}$. This notion is usually defined for locally compact groups but remains particularly relevant for permutation groups, thanks to their rich lattice of open subgroups. The next lemma is a well known fact about induced representations. In what follows, the *support* of a map $\xi\colon K\longrightarrow\mathcal{H}$ is the set $\operatorname{supp}\xi:=\{k\in K\colon\ \xi(k)\neq 0\}$.

Lemma 1.20. *Let K be a topological group and let $U\leqslant K$ be an open subgroup. Let $\sigma\colon U\curvearrowright\mathcal{K}$ be a unitary representation of U on a Hilbert space $\mathcal{K}$. Consider the representation $\pi:=\mathrm{Ind}_U^K(\sigma)$ of K, denote by $\mathcal{H}$ its underlying Hilbert space, and let*

$$\mathcal{H}_0=\{\xi\in\mathcal{H}\colon\ \operatorname{supp}\xi\subseteq U\}.$$

Then $\mathcal{H}_0$ is stable under the action of U and $\pi\big|_U\colon U\curvearrowright\mathcal{H}_0$ is isomorphic to σ.

When K is a permutation group over a countable set Ω, the richness of the lattice of open subgroups provides much control on the representation theory of K. If $\pi\colon K\curvearrowright\mathcal{H}$ is a unitary representation of K and A is a finite subset of Ω, we will write $\mathcal{H}_A$ for the closed subspace of $\mathcal{H}$ consisting of K_A-invariant vectors. That is

$$\mathcal{H}_A:=\ \{\xi\in\mathcal{H}\colon\forall k\in K_A,\ \pi(g)\xi=\xi\}.$$

If A is infinite, we set:

$$\mathcal{H}_A=\overline{\bigcup_{B\subseteq A\text{ finite}}\mathcal{H}_B}\ .$$

If $A = \{a\}$ is a singleton, we will write $\mathcal{H}_a$ instead. Finally, p_A (or p_a) will denote the orthogonal projection onto $\mathcal{H}_A$. When several representations are at play, we will specify them in the notation and write p_A^π instead.

One of the reasons closed permutation groups are a particularly well-behaved class of groups with respect to their representation theory is the fact that invariant vectors always determine the whole representation. See e.g. [Tsa12, Lem. 3.1] for a proof of this fundamental result.

Lemma 1.21. *Let K be a closed permutation group over a set Ω and let $\pi\colon K \curvearrowright \mathcal{H}$. Then:*

$$\bigcup_{A\subseteq\Omega \text{ finite}} \mathcal{H}_A \quad \text{is dense in } \mathcal{H}\,.$$

When $\mathfrak{M}$ is $\aleph_0$-categorical, these subspaces $\mathcal{H}_A$ moreover live coherently inside their underlying space. The following lemma is a consequence of the classification from [Tsa12]. More precisely, it is a slight strengthening of [JT22, Prop. 3.2]. Used on classifying sets, it is enough to recover the classification theorem of [Tsa12]. See the proof of Corollary 2.12 below for an illustration of the method.

Lemma 1.22. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and let $C \subseteq \mathfrak{M}^{\mathrm{eq}}$ be a finite subset. Let $\sigma\colon \Gamma \curvearrowright \mathcal{H}$ be a unitary representation of Γ_C. For all algebraically closed subsets A, B of $\mathfrak{M}^{\mathrm{eq}}$ containing C,*

$$\mathcal{H}_A \perp_{\mathcal{H}_{A\cap B}} \mathcal{H}_B\,,$$

i.e. $p_A p_B = p_{A\cap B}$.

Proof. Let $\pi := \mathrm{Ind}_{\Gamma_C}^{\Gamma}(\sigma)$ and let Λ be the the unitary representation $\Gamma \curvearrowright \ell^2(\mathfrak{M}^{\mathrm{eq}})$. Since $\mathfrak{M}$ is $\aleph_0$-categorical, π is isomorphic to a subrepresentation of a direct sum of copies of Λ [JT22, Fact 3.1]. Since the property we are checking is preserved under passing to a subrepresentation or forming sums, and since $\sigma \subseteq \pi\big|_G$ by Lemma 1.20, we can assume that σ is the restriction of Λ to Γ_C. It then suffices to apply [JT22, Prop. 3.2] after observing that for every subset $D \subseteq \mathfrak{M}^{\mathrm{eq}}$ containing C, we have $p_D^\sigma = p_D^\Lambda$. □

Lemma 1.23. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical, let $A \subseteq \mathfrak{M}^{\mathrm{eq}}$ be finite and let $\pi\colon \Gamma_A \curvearrowright \mathcal{H}$ be a unitary representation of Γ_A. Suppose $\xi \in \mathcal{H}$ has a finite orbit under the action of Γ_A. Then $\xi \in \mathcal{H}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)}$.*

Proof. Let $U := \{\gamma \in \Gamma_A : \gamma(\xi) = \xi\}$. By continuity of π and finiteness of $\pi(\Gamma_A)\xi$, the subgroup U of Γ is open. Using $\aleph_0$-categoricity, there exists $e \in \mathfrak{M}^{\mathrm{eq}}$ such that $U = \Gamma_e$ (see e.g. the first paragraph of [Tsa12, Sec. 5]). Then:

$$\Gamma_A \cdot e \simeq \Gamma_A/U \simeq \Gamma_A \cdot \xi.$$

Thus $\Gamma_A \cdot e$ is finite, i.e. $e \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)$. In particular, $\xi \in \mathcal{H}_e \subseteq \mathcal{H}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)}$. □

2. Non-spherical representations of G

The content of this section is devoted to classifying the non-spherical representations of G when $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Recall that $\pi\colon G \curvearrowright \mathcal{H}$ is non spherical if $\mathcal{H}_x = 0$ for every (equivalently, some) $x \in \mathbf{T}$. In Subsection 2.1, we study the classifying subtrees i.e. the definable, locally finite and algebraically closed subtrees of $\mathbf{T}^{\mathrm{eq}}$. In Subsection 2.2 we state and prove the main result of this paper, Theorem 2.13.

2.1. **The lattice of classifying subtrees.**

In order to capture the non-spherical irreducible representations of G, we start with the study of $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})$. Note that for every $X \subseteq \mathbf{T}^{\mathrm{eq}}$, any graph automorphism of $\mathbf{T}^{\mathrm{eq}}$ that fixes X also fixes the tree X generates. In particular, algebraically closed subsets of $\mathbf{T}^{\mathrm{eq}}$ are automatically trees. We will thus refer to elements of $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ as the *classifying subtrees* of $\mathbf{T}^{\mathrm{eq}}$.

Remark 2.1. Under our working assumptions, that is if $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive, $(\mathcal{C}(\mathbf{T}^{\mathrm{eq}}), \subseteq, \wedge, \vee)$ where $\wedge$ and $\vee$ are defined as in Subsection 1.3, is an ordered lattice. We postpone the proof of this fact to Corollary 2.6 below and start with some very elementary observations.

First, the smallest element of this lattice is the empty set. Next, the atoms of $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})\backslash\{\emptyset\}$ are precisely the elements of

$$\mathcal{C}_0(\mathbf{T}^{\mathrm{eq}}) := \{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)_x : x \in \mathbf{T}\}.$$

More precisely, the elements of $\mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$ are the direct successors of the empty tree and every non-empty classifying subtree of $\mathbf{T}^{\mathrm{eq}}$ contains some element of $\mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$.

Clearly, $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ and $\mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$ are stable under the action of G. Note that, because of the formalism used for defining $\mathfrak{M}^{\mathrm{eq}}$, any non-empty classifying subtree X is infinite and that each of its leaves lies in $\mathbf{T}^{\mathrm{eq}}\backslash\mathbf{T}$. Moreover, it always satisfies $X \cap \mathbf{T} \neq \emptyset$. Indeed, any graph automorphism of $\mathbf{T}^{\mathrm{eq}}$ that fixes some vertex in $\mathbf{T}^{\mathrm{eq}}\backslash\mathbf{T}$ must also fix its unique neighbor.

Proposition 2.2. *Let X be a subtree of $\mathbf{T}^{\mathrm{eq}}$ such that $X \cap \mathbf{T} \neq \emptyset$. The following equivalences hold:*

1. *X is locally finite* $\iff$ *$X \cap \mathbf{T}$ is finite and $E_X(x)$ is locally finite in $\mathfrak{M}^{\mathrm{eq}}$ for every $x \in X \cap \mathbf{T}$.*
2. *X is algebraically closed* $\iff$ *$E_X(x)$ is algebraically closed in $\mathfrak{M}^{\mathrm{eq}}$ for every $x \in X \cap \mathbf{T}$.*

Moreover, if $X \cap \mathbf{T}$ is finite:

3. *X is definable* $\iff$ *$E_X(x)$ is definable in $\mathfrak{M}^{\mathrm{eq}}$ for every $x \in X \cap \mathbf{T}$.*

Proof.

1. Assume X is locally finite. Then $X \cap \mathbf{T}$ is indeed finite since $\mathbf{T}$ is a single orbit under the action of G (Item 1 of Lemma 1.7). Let $x \in X \cap \mathbf{T}$ and let A be the subset of $\mathfrak{M}^{\mathrm{eq}}$ corresponding to $E_X(x)$. Let us show that A is finite in $\mathfrak{M}^{\mathrm{eq}}$.
   Fix $a \in \mathfrak{M}^{\mathrm{eq}}$. Then $\Gamma \cdot a \simeq G_x \cdot a_x$ by Corollary 1.10, hence:
   $$\Gamma \cdot a \cap A \simeq G_x \cdot a_x \cap V_X(x) \subseteq G \cdot a_x \cap X.$$
   Since X is locally finite, $G \cdot a_x \cap X$ is finite and so is $\Gamma \cdot a \cap A$.
   Conversely, assume that $X \cap \mathbf{T}$ is finite and that $E_X(z)$ is locally finite in $\mathfrak{M}^{\mathrm{eq}}$ for every $z \in X \cap \mathbf{T}$. Let $x \in \mathbf{T}^{\mathrm{eq}}$. Since $X \cap \mathbf{T}$ is finite $\mathbf{T}$ is G-stable, it suffices to deal with the case $x \in \mathbf{T}^{\mathrm{eq}}\backslash\mathbf{T}$. Then x is a leaf, adjacent to a unique vertex $y \in \mathbf{T}$. Setting $a := \chi(x,y)$, we have:
   $$G \cdot x \cap X = \bigcup_{z \in X \cap \mathbf{T}} G \cdot x \cap V_X(z) = \bigcup_{z \in X \cap \mathbf{T}} (\Gamma \cdot a)_z \cap V_X(z),$$
   where the last equality is given by Corollary 1.10. Recalling the assumption we made on X, $G \cdot x \cap X$ is a finite union of finite set hence finite.
2. Assume X is algebraically closed with respect to the action $G \curvearrowright \mathbf{T}^{\mathrm{eq}}$. Fix $x \in X$ and let A be the subset of $\mathfrak{M}^{\mathrm{eq}}$ corresponding to $E_X(x)$. Let us show that A is algebraically closed.
   Let $A' \subseteq A$ be finite. For every $a \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A')$, Corollary 1.10 yields:
   $$\Gamma_{A'} \cdot a \simeq G_{A'_x} \cdot a_x \,.$$
   In particular, $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A')_x \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(A'_x) \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) = X$. Thus $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A')_x \subseteq V_X(x) \cup \{x\}$ i.e. $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A') \subseteq A$.
   Conversely, assume that $E_X(x)$ is algebraically closed in $\mathfrak{M}^{\mathrm{eq}}$ for every $x \in X \cap \mathbf{T}$. Let X' be a finite subset of X. Up to replacing X' with a larger finite subset of X, we assume it is a subtree of $\mathbf{T}^{\mathrm{eq}}$ and that $X' \cap \mathbf{T} \neq \emptyset$. Let $x \in \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X')$. Let y be the unique vertex of X' witnessing $d(x,y) = d(x,X')$. We reason by induction on $n = d(x,y)$.
   - <u>$n = 0$:</u>
     In this case, $y \in X'$ and there is nothing to prove.
   - <u>$n > 0$:</u>
     Let z be the unique vertex of $\mathbf{T}^{\mathrm{eq}}$ adjacent to y that satisfies $d(z,x) = n-1$. Let A be the subset of $\mathfrak{M}^{\mathrm{eq}}$ corresponding to $E_X(y)$ and let A' be the one corresponding to

$E_{X'}(y)$. Since $G_{X'} \cdot x$ is finite and $\mathbf{T}^{\mathrm{eq}}$ is a tree, $G_{X'} \cdot z$ has to be finite too. Moreover, by Corollary 1.10,

$$G_{X'} \cdot z \simeq \Gamma_{A'} \cdot \chi(y,z).$$

Thus $\chi(y,z) \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A') \subseteq A$ i.e. $z \in X$ and we can add z to X'. The induction hypothesis now applies, yielding $x \in X$.

3. Assume that X is definable (the finiteness hypothesis is not needed for this direction). Let $x \in X \cap \mathbf{T}$ and let A be the subset of $\mathfrak{M}^{\mathrm{eq}}$ corresponding to $E_X(x)$. By Lemma 1.9, the image of $G_{\{X\}} \cap G_x$ under the restriction map $g \longmapsto g_x$ is open. Since, clearly, it is included in $\Gamma_{\{A\}}$, A is definable in $\mathfrak{M}^{\mathrm{eq}}$.

   Conversely, let X be a subtree of $\mathbf{T}^{\mathrm{eq}}$ such that $X \cap \mathbf{T}$ is finite and $E_X(x)$ is definable in $\mathfrak{M}^{\mathrm{eq}}$ for every $x \in X \cap \mathbf{T}$. Since X is a tree, we can assume $X \cap \mathbf{T}$ is non-empty. Note that:

$$G_{\{X\}} \geqslant \bigcap_{x \in X \cap \mathbf{T}} G_{\{V_X(x) \cup \{x\}\}} \,.$$

   The intersection being finite, it suffices to show that for every definable subset $A \subseteq \mathfrak{M}^{\mathrm{eq}}$, $G_{\{A_x\}}$ is open in G. Now, if A is a definable subset of $\mathfrak{M}^{\mathrm{eq}}$, then $G_{\{A_x\}}$ is the inverse image of $\Gamma_{\{A\}}$ under the restriction map $G_x \longrightarrow \Gamma$. This map being continuous by Lemma 1.9, $G_{\{A_x\}}$ is open.

□

The previous lemma directly implies the following local characterization of classifying subtrees.

Corollary 2.3. *Let X be a subtree of $\mathbf{T}^{\mathrm{eq}}$. Then:*

$$X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}}) \qquad \Longleftrightarrow \qquad X \cap \mathbf{T} \text{ is finite, non-empty, and for every } x \in X \cap \mathbf{T},\ E_X(x) \in \mathcal{C}(\mathfrak{M}^{\mathrm{eq}}).$$

Similarly to what happens with elements of $\mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ in $\mathfrak{M}^{\mathrm{eq}}$, classifying subtrees of $\mathbf{T}^{\mathrm{eq}}$ are always infinite but are algebraic closures of finite subtrees if $\mathfrak{M}$ is $\aleph_0$-categorical. In what follows, given a subtree X of $\mathbf{T}^{\mathrm{eq}}$ and subgroup H of G, we will say that X is H-stable if $H \leqslant G_{\{X\}}$.

Lemma 2.4. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical. Let X be a classifying subtree of $\mathbf{T}^{\mathrm{eq}}$. There exists a finite subtree $X_0 \subseteq X$ such that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X$. Moreover, every such X_0 satisfies $G_{X_0} \leqslant G_{\{X\}}$ with finite index. Finally, X_0 can be chosen $G_{\{X\}}$-stable, in which case G_{X_0} is a normal subgroup of $G_{\{X\}}$.*

Proof. Using Item 1 in Proposition 1.15, for every $x \in X \cap \mathbf{T}$ there exists $e^x \in \mathfrak{M}^{\mathrm{eq}}$ such that $E_X(x) \simeq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(e^x$. Define:

$$X_0 = (X \cap \mathbf{T}) \cup \{e^x_x \colon x \in X \cap \mathbf{T}\},$$

which is a finite subtree of X. Fix $x \in X \backslash X_0$. We will show that $x \in \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$.

Necessarily, x is an element of $\mathbf{T}^{\mathrm{eq}} \backslash \mathbf{T}$ hence a leaf in X. Let y be its unique neighbor, which is an element of $X \cap \mathbf{T}$. Then, using Corollary 1.10:

$$G_{X_0} \cdot x \simeq \Gamma_{e^y} \cdot \chi(x,y)\,,$$

which is finite by choice of e^y. Thus $x \in \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$ and $X \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$. The reverse inclusion holds since X is algebraically closed.

For the finite index assertion, note that for every $x \in X_0$, $G_{\{X\}} \cdot x \subseteq F_x := G \cdot x \cap X$ which is finite. Now, writing $X_0 = \{x_1, \ldots, x_n\}$ in an injective way, we have $G_{\{X\}}/G_{X_0} \simeq G_{\{X\}} \cdot (x_1, \ldots, x_n) \subseteq F_{x_1} \times \cdots \times F_{x_n}$. In particular, $G_{\{X\}}/G_{X_0}$ is finite.

For the last statement,

$$X_0' := \bigcup G_{\{X\}} \cdot X_0 \subseteq \bigcup_{1 \leqslant i \leqslant n} F_{x_i}$$

is finite, included in $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$, and stable under the action of $G_{\{X\}}$. Replace X_0 with the larger set X_0'.

□

If $\mathfrak{M}$ is transitive and has algebraicity over singletons, the algebraic closure in $\mathbf{T}$ of any edge of $\mathbf{T}$ is infinite, preventing the existence of non-empty classifying trees. However, if $\mathfrak{M}$ is $\aleph_0$-categorical, this is the only obstruction. Recall that $\langle X\rangle$ denotes the subtree of $\mathbf{T}^{\mathrm{eq}}$ generated by a subset $X \subseteq \mathbf{T}^{\mathrm{eq}}$ and that 2-transitivity if much stronger than "no algebraicity over singletons".

Proposition 2.5. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and has no algebraicity over singletons, i.e. that $\mathrm{acl}_{\mathfrak{M}}(a) = \{a\}$ for every $a \in \mathfrak{M}$. Let $X \subseteq \mathbf{T}^{\mathrm{eq}}$ be a finite subset. Then $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X)$ is a classifying subtree.*

More generally, if X is a classifying subtree of $\mathbf{T}^{\mathrm{eq}}$ and $Y \subseteq \mathbf{T}^{\mathrm{eq}}$ is either finite or a subset of a classifying subtree, $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup Y)$ is a classifying subtree.

Proof. Let X be a finite subset of $\mathbf{T}^{\mathrm{eq}}$. Since $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(\langle X\rangle)$ and $\langle X\rangle$ is finite, we may assume that X is a tree. By transitivity of $G \curvearrowright \mathbf{T}$, $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(\emptyset) = \emptyset$ and we may assume that X is non-empty. Moreover, if $X \cap \mathbf{T} = \emptyset$, then X is a singleton $\{x\}$ where $x \in \mathbf{T}^{\mathrm{eq}} \backslash \mathbf{T}$ and there exists a unique $y \in \mathbf{T}$ such that $x \sim y$. Then $G_x = G_{\{x,y\}}$ hence $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup \{y\})$ and we can assume that $X \cap \mathbf{T} \neq \emptyset$.

Claim. *After these simplifications, we have:*

$$\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T} = \bigcup_{x \in X \cap \mathbf{T}} (\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(E_X(x)) \cap \mathfrak{M})_x \ ,$$

In particular, $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T}$ is finite.

Proof of the claim. Since $\mathfrak{M}$ has no algebraicity on singletons, G_X assigns an infinite orbit to every vertex $y \in \mathbf{T}$ that such that $d(y, X) \geqslant 2$ (we are using the assumption that X is a tree). Thus, if $y \in (\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T}) \backslash X$, then $d(y, X) = 1$. Since $X \cap \mathbf{T} \neq \emptyset$, we even have $d(y, X \cap \mathbf{T}) = 1$ and there exists a unique $x \in X \cap \mathbf{T}$ such that $x \sim y$. Moreover, seeing $E_X(x)$ as a finite subset of $\mathfrak{M}^{\mathrm{eq}}$, we have by Corollary 1.10

$$G_X \cdot y \simeq \Gamma_{E_X(x)} \cdot \chi(x, y)$$

which is finite by choice of y. Thus $y \in (\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(E_X(x)) \cap \mathfrak{M})_x$. The reverse inclusion also follows from Corollary 1.10.

Finally, for every $x \in X \cap \mathbf{T}$, $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(E_X(x)) \cap \mathfrak{M}$ is finite by Items 1 and 2 of Proposition 1.15 hence so is $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T}$. $\square_{\text{claim}}$

By the above claim, we can assume that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T} \subseteq X$. Viewing things in $\mathfrak{M}^{\mathrm{eq}}$ and applying Corollary 1.10, we get that for every $x \in \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X) \cap \mathbf{T}$, $E_{\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X)}(x) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(E_X(x))$. We recall Item 1 of Proposition 1.15 and conclude with Proposition 2.3

Suppose now X is a classifying subtree and Y is a finite subset of $\mathbf{T}^{\mathrm{eq}}$. By Lemma 2.4, there exists a finite subset $X_0 \subseteq \mathbf{T}^{\mathrm{eq}}$ such that $X = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$. Then $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup Y) = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0 \cup Y)$ and the first part of the proof applies. Similarly, if Y is a subset of a classifying subtree, apply Lemma 2.4 again to get Y_0 finite such that $Y \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y_0)$. Then $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup Y) \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0 \cup Y_0)$. It is easily checked by the local characterization of Corollary 2.3 that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup Y)$ is classifying. $\square$

Corollary 2.6. *If $\mathfrak{M}$ is $\aleph_0$-categorical and has no algebraicity over singletons, $(\mathcal{C}(\mathbf{T}^{\mathrm{eq}}), \subseteq, \wedge, \vee)$ is a well-founded ordered lattice.*

Proof. If $X, Y \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$, the local characterization from Corollary 2.3 readily implies that $X \cap Y \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$. Moreover, $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X \cup Y)$ also belongs to $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ by Proposition 2.5. Thus, the ordered lattice $(\mathcal{C}(\mathbf{T}^{\mathrm{eq}}), \subseteq, \wedge, \vee)$ is well defined.

We now turn to well-foundedness. Assume $(X_n)_{n \in \mathbb{N}} \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})^{\mathbb{N}}$ is a decreasing sequence. Then $(X_n \cap \mathbf{T})_{n \in \mathbb{N}}$ is a decreasing sequence of finite subsets of $\mathbf{T}$ (Corollary 2.3) hence stationary. We can thus assume it is constant and denote by Z its unique value. By Item 3 of Proposition 1.15, the sequence $(E_{X_n}(x))_{n \in \mathbb{N}}$ is stationary for every $x \in Z$. Since Z is finite, $(X_n)_{n \in \mathbb{N}}$ itself is stationary. $\square$

2.2. **The classification theorem.**

Using the classifying subtrees, we can exhibit a family of irreducible representations of G and identify repetitions within it. The key to achieving this lies in the following lemma, which will allow us to compute invariant spaces associated with induced representations. In what follows, a representation π of a topological group H is trivial on a subgroup K if $K \subseteq \ker \pi$.

Lemma 2.7. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $X, Y \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ with X of the form $X = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$ for some $G_{\{X\}}$-stable finite subtree $X_0 \subseteq X$. Let $\sigma\colon G_{\{X\}} \curvearrowright \mathcal{K}$ be a unitary representation of $G_{\{X\}}$ that is trivial on G_{X_0}. Let $\pi := \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)$ and let $\mathcal{H}$ be the underlying Hilbert space of π. Then:*

$$\mathcal{H}_Y = \{\xi \in \mathcal{H}\colon \forall g \in G,\ [\xi(g) \neq 0 \Rightarrow gX \subseteq Y]\}.$$

In particular,

$$\mathcal{H}_X = \{\xi \in \mathcal{H}\colon\ \mathrm{supp}(\xi) \subseteq G_{\{X\}}\}$$

and p_X is the multiplication by the indicator map of $G_{\{X\}}$.

Proof. Write $\widetilde{\mathcal{H}}_Y := \{\xi \in \mathcal{H}\colon \forall g \in G,\ [\xi(g) \neq 0 \Rightarrow gX \subseteq Y]\}$. We first prove that $\mathcal{H}_Y \subseteq \widetilde{\mathcal{H}}_Y$. Let $Y' \subseteq Y$ be a finite subtree. Since $\widetilde{\mathcal{H}}_Y$ is closed, it suffices to prove that $\mathcal{H}_{Y'} \subseteq \widetilde{\mathcal{H}}_Y$.

Let $\xi \in \mathcal{H}_{Y'}$ and assume $g \in G$ is such that $\xi(g) \neq 0$. Then $\|\xi(\cdot)\|$ is constant on $G_{Y'}gG_{\{X\}}$. Since $\xi \in \ell^2(G/G_{\{X\}})$, it implies that $\left(G_{Y'}gG_{\{X\}}\right)\big/G_{\{X\}}$ is finite. By Lemma 2.4, we have $G_{X_0} \leqslant G_{\{X\}}$ with finite index thus $\left(G_{Y'}gG_{X_0}\right)\big/G_{X_0}$ is also finite. Equivalently, since X_0 is finite, every element of $g(X_0)$ has a finite orbit under the action of $G_{Y'}$. In other words, $g(X_0) \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y') \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y) = Y$.

Turning to the reverse inclusion, let $\xi \in \widetilde{\mathcal{H}}_Y$.

Claim. *We can assume that* $\mathrm{supp}\,\xi$ *is a finite union of $G_{\{X\}}$ cosets.*

Proof of the claim. Write $\mathrm{supp}\,\xi = \bigcup_{i\in I} g_i G_{\{X\}}$ without repetition for some $(g_i)_{i\in I} \in G^I$. For every $\varepsilon > 0$, there exists a finite subset $I_\varepsilon \subseteq I$ such that $\sum_{i\notin I_\varepsilon} \|\xi(g_i)\|^2 < \varepsilon$. Let $\xi_\varepsilon \in \mathcal{H}$ be the map defined by $\xi_\varepsilon(g) = \xi(g)$ if $g \in \bigcup_{i\in I_\varepsilon} g_i G_{\{X\}}$ and $\xi_\varepsilon(g) = 0$ otherwise. Then $\xi_\varepsilon \in \widetilde{\mathcal{H}}_Y$ since $\mathrm{supp}\,\xi_\varepsilon \subseteq \mathrm{supp}\,\xi$. Moreover, $\|\xi_\varepsilon - \xi\|^2 = \sum_{i\notin I_\varepsilon} \|\xi(g_i)\|^2 < \varepsilon$. Thus, $\mathcal{H}_Y$ being closed, we can replace ξ with ξ_ε. $\square_{\text{claim}}$

Next, since G_{X_0} has finite index in $G_{\{X\}}$, $\mathrm{supp}\,\xi$ is even a finite union of G_{X_0}-cosets and we can write:

$$\mathrm{supp}(\xi) = \bigcup_{1\leqslant i\leqslant n} g_i G_{X_0}$$

where $g_1, \ldots, g_n \in G$.

Let $Y_0 \subseteq Y$ be as in Lemma 2.4 and define:

$$Y' = Y_0 \cup \bigcup_{1\leqslant i\leqslant n} g_i(X')$$

which is a finite subset of Y . Indeed, recall that $g_1, \ldots, g_n$ belong to $\mathrm{supp}(\xi)$ hence satisfy $g_i(X) \subseteq Y$ for every $i \leqslant n$. Let us show that ξ is $G_{Y'}$-stable. To that aim, fix $u \in G_{Y'}$ and $g \in G$.

- <u>If $g \notin g_1 G_{X_0} \cup \cdots \cup g_n G_{X_0}$:</u>

  Then $u^{-1}g \notin u^{-1}g_1 G_{X_0} \cup \cdots \cup u^{-1}g_n G_{X_0}$. Since for every $i \leqslant n$, $g_i(X_0) \subseteq Y'$, it follows that $u^{-1}g_i G_{X_0} = g_i G_{X_0}$ hence $u^{-1}g \notin \mathrm{supp}\,\xi$. In other words:

  $$(u \cdot \xi)(g) = \xi(u^{-1}g) = 0 = \xi(g).$$

- <u>If $g \in g_i G_{X_0}$ for some $i \leqslant n$:</u>

  In this case, $u^{-1}g \in G_{Y'}g = gG_{g^{-1}(Y')}$. Since $g(X_0) = g_i(X_0) \subseteq Y'$, we have $G_{g^{-1}(Y')} \subseteq G_{X_0}$. We can thus pick $v \in G_{X_0}$ such that $u^{-1}g = gv$. Then:

  $$(u \cdot \xi)(g) = \xi(u^{-1}g) = \xi(gv) = \sigma(v^{-1})(\xi(g)) = \xi(g)$$

  where the last equality follows from the assumption that σ is trivial on G_{X_0}.

As for the last statement, assume $X = Y$ and let $g \in G$ be such that $g(X) \subseteq X$. It suffices to show that $X \subseteq g(X)$. Let $x \in X$. Since X is locally finite, $E := G \cdot x \cap X$ is finite. Now, $g(E) \subseteq E$ hence $g(E) = E$ by cardinality and $x \in g(X)$.

□

Recall that if H, H' are conjugate subgroups of G, say $gHg^{-1} = H'$ for some $g \in G$, and if τ is a representation of H' on a Hilbert space $\mathcal{K}$, then $\tau^g \colon H \curvearrowright \mathcal{K}$ is the representation of H given by $\tau^g(h) = \tau(ghg^{-1})$ for every $h \in H$. The *trivial representation* of a topological group is its trivial action on $\mathbb{C}$. A vector in a representation $\pi \colon G \curvearrowright \mathcal{H}$ is G-cyclic if its orbit under the action of G spans $\mathcal{H}$ as a Hilbert space. A subspace $\mathcal{K} \subseteq \mathcal{H}$ is G-cyclic if $\bigcup\{\pi(g)\mathcal{K} \colon g \in G\}$ spans $\mathcal{H}$ as a Hilbert space.

Proposition 2.8. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ be of the form $X = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$ for some $G_{\{X\}}$-stable finite subtree $X_0 \subseteq X$. Let σ be an irreducible unitary representation of $G_{\{X\}}$ that is trivial on G_{X_0}. Then $\pi := \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)$ is irreducible. It is spherical exactly when either X is empty or both $X \in \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$ and σ is the trivial representation.*

Moreover, suppose Y is another classifying subtree of $\mathbf{T}^{\mathrm{eq}}$ of the form $Y = \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y_0)$ for some $G_{\{Y\}}$-stable finite subtree $Y_0 \subseteq Y$. If τ is an irreducible representation of $G_{\{Y\}}$ which is trivial on G_{Y_0} and such that $\pi' := \mathrm{Ind}_{G_{\{Y\}}}^{G}(\tau)$ is isomorphic to π, there exists $g \in G$ such that $g(X) = Y$ and $\sigma \simeq \tau^g$.

Proof. We first prove the irreducibility of π by showing that every non-zero vector in $\mathcal{H}$ is G-cyclic. Indeed, let $\xi \in \mathcal{H}\backslash\{0\}$. There exists $g \in G$ such that $\xi(g) \neq 0$. Up to replacing with ξ with $\pi(g)\xi$, we can assume that $\xi(1_G) \neq 0$. By Lemma 2.7, $p_X(\xi) \neq 0$. Now, by Lemma 1.19 and the Alaoglu-Birkhoff Theorem 1.18, $p_X(\xi)$ lies in the G-cyclic hull of ξ. It thus suffices to prove the following claim:

Claim. *Every non-zero vector of $\mathcal{H}_X$ is G-cyclic in $\mathcal{H}$.*

Proof of the claim. Using Lemmas 2.7 and 1.20, the representation $\pi\big|_{G_{\{X\}}} \colon G_{\{X\}} \curvearrowright \mathcal{H}_X$ is isomorphic to σ hence irreducible. Thus, every non-zero vector of $\mathcal{H}_X$ is $G_{\{X\}}$-cyclic in $\mathcal{H}_X$, a space which itself is G-cyclic in $\mathcal{H}$. □claim

Suppose π is spherical and let $x \in \mathbf{T}$. Write $A := \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. Then $\mathcal{H}_x \subseteq \mathcal{H}_{A_x}$. Thus, given $\xi \in \mathcal{H}_x\backslash\{0\}$ and $g \in G$ such that $\xi(g) \neq 0$, we have $g(X) \subseteq A_x$ by Lemma 2.7. Now, either X is empty or $X \in \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$. Assume the latter and, without loss of generality, that $x \in X$, i.e. that $X = A_x$. Then $G_{\{X\}} = G_x$ and $\xi(1_G)$ is a non-zero invariant vectors of σ. Since σ is irreducible, it must be trivial. Conversely, inducing the trivial representation of G_x to G yields the action $G \curvearrowright \ell^2(G/G_x)$ which is indeed spherical. In the case $X = \emptyset$, σ is automatically trivial and $G_{\{X\}} = GG$. In this case, inducing does nothing hence π is the trivial representation of G which is indeed spherical.

Next, assume π and π' are isomorphic and denote by $\mathcal{H}$ and $\mathcal{H}'$ their respective underlying Hilbert space. Then $\mathcal{H}'_Y \simeq \mathcal{H}_Y \neq 0$. Let $\xi \in \mathcal{H}_Y\backslash\{0\}$ and $g \in G$ be such that $\xi(g) \neq 0$. By Lemma 2.7, g satisfies $g(X) \subseteq Y$. By symmetry, we can find $h \in G$ such that $h(Y) \subseteq X$. Since X and Y are locally finite, this implies $g(X) = Y$. In particular, $gG_{\{X\}}g^{-1} = G_{\{Y\}}$. Finally, using Lemmas 2.7 and 1.20 again for the first and last isomorphisms:

$$\sigma \simeq \left(G_{\{X\}} \overset{\pi}{\curvearrowright} \mathcal{H}_X\right) \simeq \left(G_{\{X\}} \overset{\pi'}{\curvearrowright} \mathcal{H}'_X\right) \simeq \left(G_{\{X\}} \overset{\pi'^g}{\curvearrowright} \mathcal{H}'_Y\right) = \left(G_{\{Y\}} \overset{\pi'}{\curvearrowright} \mathcal{H}'_Y\right)^g \simeq \tau^g .$$

The second isomorphism is the restriction of the one between π and π'. The third is $\pi(g)$.

□

In order to complete the classification, we need to show that every non-spherical irreducible representation of G is of the above form. To do so, we prove an analogue of [Ol'80, Lem. 5.1]. We deal with the most elementary case first, regarding elements of $\mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$.

Lemma 2.9. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $\pi \colon G \curvearrowright \mathcal{H}$ be a unitary representation of G. Let $A := \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$. For every $x \neq y \in \mathbf{T}$,*

$$\mathcal{H}_{A_x} \perp_{\mathcal{H}_x} \mathcal{H}_{A_y}.$$

Proof. Let $x \neq y \in \mathbf{T}$. Let $A' \subseteq A$ be finite. By Lemma 1.19, it suffices to show that $p_{A'_x} p_x p_{A'_y} = p_{A'_x} p_{A'_y}$, i.e. that for all $\xi \in H_{A'_x}$ and $\eta \in \mathcal{H}_{A'_x}$,

$$\langle \xi, p_x \eta \rangle = \langle \xi, \eta \rangle.$$

Let $\varepsilon > 0$. By the Alaoglu-Birkhoff Theorem 1.18, there exists $g_1, \dots, g_n \in G_x$ and $\lambda_1, \dots, \lambda_n \geqslant 0$ such that $\lambda_1 + \cdots + \lambda_n = 1$ and:

$$\|p_x \eta - \sum_i \lambda_i \pi(g_i)\eta\| < \varepsilon.$$

Claim. *Let $X := A'_x$ and $Y := \langle A'_y \cup \{x\} \rangle$. Then:*

$$G_x = G_X G_Y$$

Proof of the claim. Let $g \in G_x$. Let z be the unique neighbor of x in $\mathbf{T}$ such that $d(x,y) = d(z,y) + 1$ and let $a := \chi(x,z)$. We first show that $\Gamma = \Gamma_{A'} \Gamma_a$.

Let $A'' := \bigcup \Gamma \cdot A'$ which is a finite subset of $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)$ that is stable under the action of Γ. Thus $\Gamma_{A''} \leqslant \Gamma_{A'} \leqslant \Gamma$ and $\Gamma_{A''}$ is a finite-index normal subgroup of Γ. Consider now the equivalence relation $\sim$ on $\mathfrak{M}$ given by $b \sim b'$ if $\Gamma_{A''} \cdot b \sim \Gamma_{A''} \cdot b'$. Then $\sim$ is Γ-invariant by normality of $\Gamma_{A''}$. Because $\mathfrak{M}$ is 2-transitive, the action $\Gamma \curvearrowright \mathfrak{M}$ is primitive hence $\sim$ is trivial: either $\sim$ is equality or $b \sim b'$ for every $b, b' \in \mathfrak{M}$. Since $\Gamma_{A''}$ has finite index in Γ, the relation $\sim$ has only finitely many equivalence classes and the first case is excluded. We have thus shown that the action $\Gamma_{A''} \curvearrowright \mathfrak{M}$ is transitive. A fortiori, so is $\Gamma_{A'} \curvearrowright \mathfrak{M}$. Now, given $\gamma \in \Gamma$, there exists $\delta \in \Gamma_{A'}$ such that $\delta(a) = \gamma(a)$. In other words, $\delta^{-1}\gamma \in \Gamma_a$ hence $\gamma \in \delta \Gamma_a \subseteq \Gamma_{A'} \Gamma_a$.

Now, by Lemma 1.9, the restriction morphism $r_x \colon h \longmapsto h_x$ satisfies $r_x(G_X) = \Gamma_{A'}$ and $r_x(G_Y) = \Gamma_a$. Thus, $g_x \in \Gamma = \Gamma_{A'} \Gamma_a = r_x(G_X) r_x(G_Y)$ and there exist $h \in G_X$ and $k \in G_Y$ such that $(hk)_x = g_x$. Visibly, $gk^{-1} \in G_X$ thus $g \in G_X k \subseteq G_X G_Y$. $\square_{\text{claim}}$

By the claim, there exist $h_1, \dots, h_n \in G_X$ and $k_1, \dots, k_n \in G_Y$ such that for every $i \leqslant n$, $g_i = h_i k_i$. It follows that $\langle \xi, \pi(g_i)\eta \rangle = \langle \pi(h_i^{-1})\xi, \pi(k_i)\eta \rangle = \langle \xi, \eta \rangle$ and:

$$|\langle \xi, p_x \eta \rangle - \langle \xi, \eta \rangle| = |\langle \xi, p_x \eta \rangle - \sum_i \lambda_i \langle \xi, \pi(g_i)\eta \rangle| = |\langle \xi, p_x \eta - \sum_i \lambda_i \pi(g_i)\eta \rangle| \leqslant \|\xi\| \cdot \varepsilon$$

Since ε is arbitrary, the result follows. $\square$

In our proof of a similar statement for elements of $\mathcal{C}(\mathbf{T}^{\mathrm{eq}}) \backslash \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$, the following structure will appear:

$$\mathfrak{N} := \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(V_{\mathbf{T}^{\mathrm{eq}}}(z))$$

where z is a vertex of $\mathbf{T}$. The structure $\mathfrak{N}$ is represented in Figure 2. The group G_z acts on $\mathfrak{N}$ hence induces a permutation group $\Delta \leqslant \mathrm{Sym}(\mathfrak{N})$. It easily checked that

$$\Delta \simeq \Gamma \ltimes \prod_{a \in \mathfrak{M}} \Gamma_a / \Gamma_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)}.$$

In particular, Δ is Roelcke-precompact if $\mathfrak{M}$ is $\aleph_0$-categorical [Tsa12, Prop. 2.2]. The following essentially shows that $\mathfrak{N}$ weakly eliminates imaginaries.

Proposition 2.10. *Let C be a finite subtree of $\mathfrak{N}$ and let $\sigma := \Delta_C \curvearrowright \mathcal{H}$ be a unitary representation of Δ_C. For every algebraically closed subsets $A, B \subseteq \mathfrak{N}$ containing C, we have:*

$$\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B,$$

i.e. $p_A p_B = p_{A \cap B}$.

Proof. Exactly as in Lemma 1.22, we can assume that $C = \emptyset$ up to replacing σ with $\mathrm{Ind}_{\Delta_C}^{\Delta}(\sigma)$. Since Δ is Roelcke precompact, [JT22, Prop. 3.2] applies hence it suffices to prove the following.

Claim. *Let $X \subseteq \mathfrak{N}$ be a finite subtree of $\mathfrak{N}$ and let $\xi \in \mathcal{H}$ be such that $\Delta_X \cdot \xi$ is finite. Then $\xi \in \mathcal{H}_{\mathrm{acl}_{\mathfrak{N}}(X)}$.*

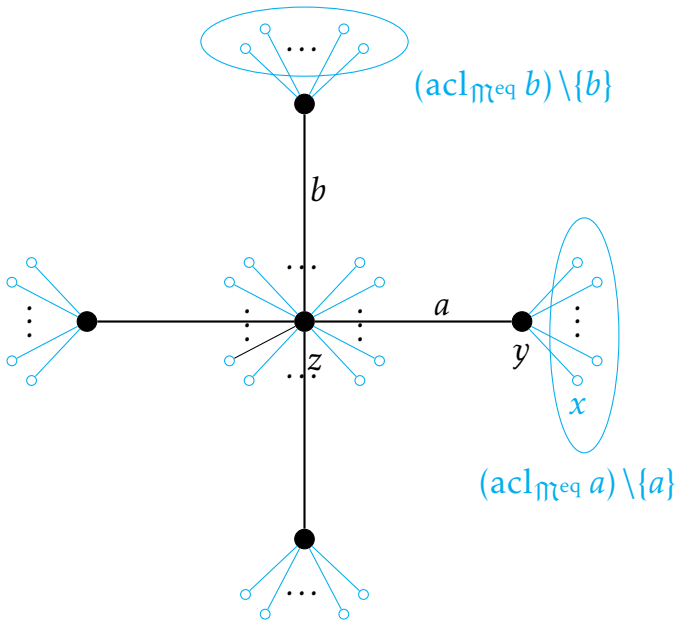

FIGURE 2. The structure $\mathfrak{N}$.

Proof of the claim. Note that we can replace X any other finite subtree of $\mathfrak{N}$ that has the same algebraic closure in $\mathfrak{N}$. In particular, since $\Delta_X = \Delta_{\langle X\cup\{z\}\rangle}$, we can assume that $z \in X$. In this case, $\mathrm{acl}_{\mathfrak{N}}(X) = \mathrm{acl}_{\mathfrak{N}}(X \cap (B_{\mathbf{T}^{\mathrm{eq}}}(z,1))$ hence we can assume that X is contained in $B_{\mathbf{T}^{\mathrm{eq}}}(z,1)$. Let $U := \{\delta \in \Delta\colon \sigma(\delta)\xi = \xi\}$. By continuity of σ and finiteness of $\sigma(\Delta_X)\xi$, U is open. Thus, there exists a finite subtree Y of $\mathfrak{N}$ such that $X \subseteq Y$ and $\Delta_Y \subseteq U$.

Using Lemma 1.3, we will denote by $\iota\colon \Gamma \longrightarrow \Delta$ the continuous group morphism that sends γ to the unique element of Δ for which every local action is γ. Let $A := E_X(z)$ be seen as a finite subset of $\mathfrak{M}^{\mathrm{eq}}$. Since $X \subseteq B_{\mathbf{T}^{\mathrm{eq}}}(z,1)$, we have $\iota(\Gamma_A) \subseteq \Delta_X$. Thus, setting $\sigma_z := \sigma \circ \iota$, the action $\Gamma\colon \sigma_z \curvearrowright \mathcal{H}$ is a unitary representation of Γ such that $\sigma_z(\Gamma_A)\cdot\xi$ is finite. By Lemma 1.23, there exists $e \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)$ such that $\iota(\Gamma_{A\cup\{e\}}) \subseteq U$.

Similarly, for every $y \in V_X(z) \cap \mathbf{T}$, we will denote by $\iota_y\colon \Gamma_a \longrightarrow \Delta$ the continuous group morphism that sends γ to the unique element of Δ for which every local action is $\mathrm{id}_{\mathfrak{M}^{\mathrm{eq}}}$, except at y where it is γ. Since $X \subseteq B_{\mathbf{T}^{\mathrm{eq}}}(z,1)$, we have $\iota_y(\Gamma_a) \subseteq \Delta_X$ for every $y \in V_X(z) \cap \mathbf{T}$. Again, this gives rise to representations $\sigma_y\colon \Gamma_a \curvearrowright \mathcal{H}$ with $\sigma_y(\Gamma_a)\cdot\xi$ finite for every $y \in V_X(z) \cap \mathbf{T}$. Applying Lemma 1.23, there exists $e^y \in \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$ such that $\iota_y(\Gamma_{a,e^y}) \subseteq U$.

We now define the following finite subtree of $\mathfrak{N}$:

$$X' := X \cup \{e_z\} \cup \left\{e_z^y\colon y \in V_X(z) \cap \mathbf{T}\right\} \cup \left\{e_y^y\colon y \in V_X(z) \cap \mathbf{T}\right\} \cup (Y \cap \mathrm{acl}_{\mathfrak{N}}(X)).$$

Recalling Corollary 1.10, we have $\mathrm{acl}_{\mathfrak{N}}(X') \subseteq \mathrm{acl}_{\mathfrak{N}}(X)$. We now show that $\xi \in \mathcal{H}_{X'}$, which will complete the proof of the claim.

Let $g \in \Delta_{X'}$. Fixing $y \in V_{X'}(z) \cap \mathbf{T}$, we have $g_y^{-1} g_z \in \Gamma_{a,e^y}$ hence $u := \iota_y(g_y^{-1} g_z) \in U$. Thus, replacing g with gu, we can assume that $g_y = g_z$ without altering any other local action of g. Since X' is finite, we will in fact assume that $g_y = g_z$ for every $y \in V_{X'}(z) \cap \mathbf{T}$. Next, using Corollary 1.4, let v be the unique element of Δ such that:

- $v(z) = z$,
- $v_z = \mathrm{id}_{\mathfrak{M}^{\mathrm{eq}}}$,
- $v_y = \mathrm{id}_{\mathfrak{M}^{\mathrm{eq}}}$ for every $y \in V_Y(z) \cap \mathbf{T}$,
- $v_y = g_y^{-1} g_z$ for every $y \in V_{\mathbf{T}}(z) \setminus Y$.

Visibly, v fixes Y hence $v \in U$. Thus, replacing g with gv, we can assume that $g_y = g_z$ for every $y \in V_{\mathbf{T}}(z) \setminus Y$. Note that the local actions at elements of $X' \cap \mathbf{T}$ are left untouched.

Finally, recall that $Y \setminus X' \subseteq \mathfrak{N} \setminus \mathrm{acl}_{\mathfrak{N}}(X')$. Thus, Neumann's Lemma [BBMN76, Th. 1] provides $\gamma \in \Gamma_{E_{X'}(z)}$ such that $\gamma(E_Y(z)) \cap E_Y(z) = E_{X'}(z)$. Denoting $w := \iota(\gamma)$, we have:

- $(gw)_z = g_{w(z)} w_z = g_z \gamma$,
- for every $y \in V_{X'}(z) \cap \mathbf{T}$, $(gw)_y = g_{w(y)} w_y = g_y \gamma = g_z \gamma$,
- for every $y \in V_Y(z) \cap \mathbf{T} \setminus X$, we have $w(y) \notin Y$ hence $(gw)_y = g_{w(y)} w_y = g_z \gamma$.

Since $\gamma \in \Gamma_{A \cup \{e\}}$, $w = \iota(\gamma) \in U$. Thus, after replacing g with gw, we have $g_y = g_z$ for every $y \in Y \cap \mathbf{T}$. Visibly, $\iota(g_z)^{-1} g \in \Delta_Y$ hence $g \in U \Delta_Y \subseteq U$ i.e. g fixes ξ. $\square_{\text{claim}}$

$\square$

The following result is inspired by [Ol'80, Lem. 5.1].

Lemma 2.11. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}}) \setminus \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$ and let $Y \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ be such that $X \not\subseteq Y$.*

There exists a proper classifying subtree $X' \subseteq X$ such that for every representation $\pi \colon G \curvearrowright \mathcal{H}$ of G, we have $p_Y p_X = p_Y p_{X'}$.

Proof. First, note that we can always replace Y with a larger classifying subtree. Indeed, suppose $Y' \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ is such that $Y \subseteq Y'$. Suppose also that there exists $X' \subseteq X$ such that for every representation $\pi \colon G \curvearrowright \mathcal{H}$ of G, we have $p_{Y'} p_X = p_{Y'} p_{X'}$. Since $Y \subseteq Y'$ we have $p_Y p_{Y'} = p_Y$ and:

$$p_Y p_X = p_Y p_{Y'} p_X = p_Y p_{Y'} p_{X'} = p_Y p_{X'}.$$

Now, since $X \cap \mathbf{T}$ is finite by Corollary 2.3, X has finite diameter and there exists $x \in X \setminus Y$ at maximal distance from Y (note that the existence of Y implies that X is non-empty). Then x is a leaf in X. Since X algebraically closed, we have $x \in \mathbf{T}^{\mathrm{eq}} \setminus \mathbf{T}$ (recall Remark 2.1). In particular, there exists a unique $y \in X \cap \mathbf{T}$ such that $x \sim y$. We distinguish two cases.

Case 1: $y \in Y$.

Define the subtree $X' \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ as follows:

$$X' := X \setminus [V_X(y) \setminus Y] \,.$$

In other words, X' is obtained from X by replacing $E_X(y)$ with $E_X(y) \cap E_Y(y)$. All of the vertices so removed are leaves in X by maximality of $d(x, Y)$, hence X' is a tree. Moreover, every $x' \in X' \setminus \{y\}$ satisfies $E_{X'}(x') = E_X(x')$. Thus $X' \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ by Corollary 2.3. It is strictly contained in X since $x \notin X'$.

Claim. *Up to replacing Y with a larger classifying tree, we can assume that $X' = X \cap Y$.*

Proof of the claim. Let $Y' := \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup X')$. By Proposition 2.5, this is a classifying subtree of $\mathbf{T}^{\mathrm{eq}}$. Moreover, applying Corollary 1.10 to finite subtrees of $Y \cup X'$ at y yields $E_{Y'}(y) = E_Y(y)$. In particular, we still have $x \notin Y'$. $\square_{\text{claim}}$

We are thus left with the following simplified situation, illustrated in Figure 3: $X \setminus Y$ is only comprised of neighbors of y. Note that the elements of $X \setminus Y$ are all leaves in X hence elements of $\mathbf{T}^{\mathrm{eq}} \setminus \mathbf{T}$.

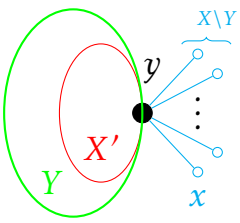

Figure 3. Situation after simplifications.

Using Lemma 2.4, let X_0, Y_0 and X_0' be finite subtrees of X, Y and X' respectively such that:

$$\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X, \quad \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y_0) = Y \quad \text{and} \quad \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0') = X'.$$

Up to replacing X_0, Y_0, X_0' with larger finite trees, we can assume that $X_0 \cap Y_0 = X_0'$ and that $y \in X_0'$. Let $A := E_{X_0}(y)$, $B := E_{Y_0}(y)$ and $C := E_{X_0'}(y)$, seen as finite subsets of $\mathfrak{M}^{\mathrm{eq}}$. Necessarily, $A \cap B = C$. Moreover, we can assume that A and B are large enough so that $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A) = E_X(y)$, $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B) = E_Y(y)$ and $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(C) = E_{X'}(y)$.

Define:

$$Z := X_0' \cup (\mathfrak{M}^{\mathrm{eq}})_y = X_0 \cup (\mathfrak{M}^{\mathrm{eq}})_y \,.$$

Then $G_{X_0'}$ leaves Z stable as a set hence acts on $\mathcal{H}_Z$. Since the action $G_{X_0'} \curvearrowright \mathcal{H}_Z$ is trivial on G_Z, it factorizes into an action of the quotient group $G_{X_0'}/G_Z$. Now, $G_{X_0'}/G_Z$ identifies with Γ_C through Lemma 1.9. Factorizing and restricting π thus gives rise to the following representation:

$$\sigma \colon \Gamma_C \curvearrowright \mathcal{K} := \mathcal{H}_Z.$$

Similarly, $G_{X_0}/G_Z \simeq \Gamma_A$ and $G_{Y_0}/(G_{Y_0} \cap G_Z) \simeq \Gamma_B$, both subgroups of Γ_C. Recalling that $\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A) \cap \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(C)$, Lemma 1.22 applied to the representation $\sigma \colon \Gamma_C \curvearrowright \mathcal{K}$ yields:

$$p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)} p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)} = p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(C)}$$

Claim. *Projections associated with σ identify with restrictions of projections associated with π as follows:*

$$p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)} = p^{\pi}_X\big|_{\mathcal{H}_Z}, \quad p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)} = p^{\pi}_Y\big|_{\mathcal{H}_Z} \quad \& \quad p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(C)} = p^{\pi}_{X'}\big|_{\mathcal{H}_Z}.$$

Proof of the claim. We only prove it for Y, the other identifications are similar and more straightforward. First, note that:

$$p^{\pi}_Y \mathcal{H}_Z = \mathcal{H}_Y \cap \mathcal{H}_Z.$$

Indeed, let $\xi \in \mathcal{H}_Z$ and $\varepsilon > 0$. By Lemma 1.19, there exists $Y_1 \subseteq Y$ finite such that:

$$\|p^{\pi}_{Y_1}\xi - p^{\pi}_Y \xi\| \leqslant \varepsilon.$$

Up to replacing Y_1 with a larger finite subset of Y, we can assume $X_0' \subseteq Y_1$. Then G_{Y_1} leaves Z, and hence $\mathcal{H}_Z$, stable. In particular, using the Alaoglu-Birkhoff Theorem 1.18 and the fact that $\mathcal{H}_Z$ is closed, we get:

$$p^{\pi}_{Y_1}\xi \in \mathcal{H}_Z.$$

Since ε was arbitrarily small and $\mathcal{H}_Z$ is closed, $p^{\pi}_Y \xi \in \mathcal{H}_Z$. The other inclusion is trivial.

Now, $p^{\pi}_Y\big|_{\mathcal{H}_Z}$ and $p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)}$ both are orthogonal projections defined on $\mathcal{H}_Z$. To prove they are equal, it suffices to show they have the same image, i.e. that $\mathcal{H}_Y \cap \mathcal{H}_Z = \mathcal{K}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)}$.

Let $\xi \in \mathcal{H}_Y \cap \mathcal{H}_Z$ and $\varepsilon > 0$. As above, there exists Y_1 a finite subset of Y containing X_0' such that $\|p^{\pi}_{Y_1}\xi - \xi\| \leqslant \varepsilon$ and $p^{\pi}_{Y_1}\mathcal{H}_Z \subseteq \mathcal{H}_Z$. Let $B_1 := E_{Y_1}(y)$ seen as a finite subset of $\mathfrak{M}^{\mathrm{eq}}$. Let $\gamma \in \Gamma_{B_1}$. Using surjectivity in Lemma 1.9, pick $g \in G_{Y_1}$ such that $g_y = \gamma$. Then:

$$\sigma(\gamma) p^{\pi}_{Y_1}(\xi) = \pi(g) p^{\pi}_{Y_1}\xi = p^{\pi}_{Y_1}\xi,$$

hence $p^{\pi}_{Y_1}(\xi) \in \mathcal{K}_{B_1} \subseteq \mathcal{K}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)}$. Since ε was arbitrarily small and $\mathcal{K}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)}$ is closed, $\xi \in \mathcal{K}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)}$.

Conversely, let $B_1 \subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)$ be finite. Let $Y_1 := X_0' \cup (B_1)_y$. Clearly, $\mathcal{K}_{B_1} \subseteq \mathcal{H}_Z \cap \mathcal{H}_{Y_1} \subseteq \mathcal{H}_Z \cap \mathcal{H}_Y$. Since B_1 was arbitrary and $\mathcal{H}_Z \cap \mathcal{H}_Y$ is closed, $\mathcal{K}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)} \subseteq \mathcal{H}_Z \cap \mathcal{H}_Y$.

$\square_{\text{claim}}$

By the previous claim, and because $X, X' \subseteq Z$:

$$p^{\pi}_Y p^{\pi}_X = p^{\pi}_Y p^{\pi}_X p^{\pi}_Z = p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(B)} p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(A)} p^{\pi}_Z = p^{\sigma}_{\mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(C)} p^{\sigma}_Z = p^{\pi}_{X'} p^{\pi}_Z = p^{\pi}_{X'}$$

which completes the proof of Case 1. Indeed, $p^{\pi}_{X'} = p_Y p^{\pi}_{X'}$ since $X' \subseteq Y$.

Case 2: $y \notin Y$.

Let $z \in \mathbf{T}$ be the unique neighbor of y such that $d(z, Y) = d(y, Y) - 1$. Note that $y \notin \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup \{z\})$ by 2-transitivity of $\mathfrak{M}$. Thus, up to replacing Y with $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup \{z\})$, we can assume that $z \in Y$. Case 1 being settled allows us to make the following two additional assumptions.

Claim. *We can assume that:*

1. $z \in X$.
2. *for every* $y' \in (V_X(z) \setminus Y) \cap \mathbf{T}$, $E_X(y') = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\chi(z, y'))$.

Proof of the claim.

1. Suppose that $z \notin X$. Since $d(x, Y)$ is maximal, X must be contained in $(\mathfrak{M}^{\mathrm{eq}})_y$. More precisely, there $A \in \mathcal{C}(\mathfrak{M}^{\mathrm{eq}})$ such that $X = A_y$. Since X itself is algebraically closed, we must have $X \cap \mathbf{T} = \{y\}$ i.e. $A \cap \mathfrak{M} = \emptyset$. Writing $a := \chi(y, z)$, we distinguish two cases:
   - If $A \not\subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$, replace Y with $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup \{y\})$ and x with any element of $(A \setminus \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a))_y$. We are now back into Case 1.
   - If $A \subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a)$, then in fact $A \subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(a) \setminus \{a\}$ since $A \cap \mathfrak{M} = \emptyset$. By Corollary 1.17, $A = \mathrm{acl}\, \mathfrak{M}^{\mathrm{eq}}(\emptyset)$ hence $X \in \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$, a contradiction.
2. Suppose that there exists $y' \in (V_X(z) \setminus Y) \cap \mathbf{T}$ such that:
$$E_X(y') \not\subseteq \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\chi(z, y'))\,.$$
   Replacing y with y' and x with any $x' \in V_X(y') \setminus \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\chi(z, y'))_{y'}$, we can assume that $y = y'$. Finally, replacing Y with $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup \{y'\})$, we are again back into Case 1.

$\square_{\mathrm{claim}}$

We now assume the statements in previous claim and define X' as follows:
$$X' := X \setminus (B_{\mathbf{T}^{\mathrm{eq}}}(z, 2) \setminus Y)$$
In other words, X' is the proper subtree of $\mathbf{T}^{\mathrm{eq}}$ obtained from X by replacing $V_X(z)$ with $V_X(z) \cap V_Y(z)$ and discarding the disconnected vertices. Since $x, y \notin X'$, this is a proper subtree of X. What is more, it follows from Corollary 2.3 that $X' \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$. Up to replacing Y with $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y \cup X')$, we will assume that that $X' = X \cap Y$ (this preserves the fact that $x, y \notin Y$).

We now conclude similarly as in Case 1. Using Lemma 2.4, let X_0, Y_0 and X_0' be finite subtrees of X, Y and X' respectively such that:
$$\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X, \quad \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y_0) = Y \quad \text{and} \quad \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0') = X'.$$
Up to replacing X_0, Y_0, X_0' with larger finite trees, we can assume that $X_0 \cap Y_0 = X_0'$ and that $y, z \in X_0'$. Define:
$$Z := X_0 \cup \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(V_{\mathbf{T}^{\mathrm{eq}}}(z)).$$

The situation is slightly more complicated than that of Case 1. Instead of seeing the structure $\mathfrak{M}^{\mathrm{eq}}$ around y, we see the larger structure $\mathfrak{N} := \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(V_{\mathbf{T}^{\mathrm{eq}}}(z))$ around z. Let $A := X_0 \cap \mathfrak{N}$, $B := Y_0 \cap \mathfrak{N}$ and $C := Y_0 \cap \mathfrak{N}$. Necessarily, $A \cap B = C$. Moreover, we can assume that A and B are large enough so that $\mathrm{acl}_{\mathfrak{N}}(A) = X \cap \mathfrak{N}$, $\mathrm{acl}_{\mathfrak{N}}(B) = Y \cap \mathfrak{N}$ and $\mathrm{acl}_{\mathfrak{N}}(C) = X' \cap \mathfrak{N}$.

Then, $G_{X_0'}$ leaves Z stable as a set hence acts on $\mathcal{H}_Z$. Again, this representation is trivial on G_Z hence factorizes into a representation of $G_{X_0'}/G_Z$. The restriction morphism $G_z \longrightarrow \Delta$ sends G_{X_0}, G_{Y_0} and $G_{X_0'}$ onto Δ_A, Δ_B and Δ_C respectively. What is more, it factorizes into topological group isomorphisms $G_{X_0}/G_Z \simeq \Delta_A$, $G_{Y_0}/(G_{Y_0} \cap G_Z) \simeq \Delta_B$ and $G_{X_0'}/G_Z \simeq \Delta_C$.

Factorizing and restricting π gives rise to the following representation:
$$\sigma \colon \Delta_C \curvearrowright \mathcal{K} := \mathcal{H}_Z.$$
Recalling that $\mathrm{acl}_{\mathfrak{N}}(A) \cap \mathrm{acl}_{\mathfrak{N}}(B) = \mathrm{acl}_{\mathfrak{N}}(C)$, Proposition 2.10 applied to σ yields:
$$p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(B)} p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(A)} = p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(C)}$$
Just as in Case 1 and with the same proof,
$$p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(A)} = p^{\pi}_X\big|_{\mathcal{H}_Z}\,, \quad p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(B)} = p^{\pi}_Y\big|_{\mathcal{H}_Z} \quad \& \quad p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(C)} = p^{\pi}_{X'}\big|_{\mathcal{H}_Z}.$$
Hence:
$$p^{\pi}_Y p^{\pi}_X = p^{\pi}_Y p^{\pi}_X p^{\pi}_Z = p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(B)} p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(A)} p^{\pi}_Z = p^{\sigma}_{\mathrm{acl}_{\mathfrak{N}}(C)} p^{\sigma}_Z = p^{\pi}_{X'} p^{\pi}_Z = p^{\pi}_{X'}$$
which completes the proof of Case 2. Indeed, $p^{\pi}_{X'} = p_Y p^{\pi}_{X'}$ since $X' \subseteq Y$.

$\square$

The previous results can be used to find irreducible induced representations in any non-spherical representation of G.

Corollary 2.12. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Let $\pi\colon G \curvearrowright \mathcal{H}$ be a non-zero and non-spherical unitary representation of G. There exists:*

- *a classifying subtree $X \subseteq \mathbf{T}^{\mathrm{eq}}$,*
- *a finite $G_{\{X\}}$-stable subtree X_0 of X such that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X$,*
- *an irreducible representation σ of the finite group $G_{\{X\}}/G_{X_0}$*

such that $\mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma) \subseteq \pi$.

Proof. By Lemma 1.21, there exists a finite subtree $Y \subseteq \mathbf{T}$ such that $\mathcal{H}_Y \neq 0$. Then $Z := \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Y)$ is a classifying subtree of $\mathbf{T}^{\mathrm{eq}}$ by Proposition 2.5 such that $\mathcal{H}_Z \supseteq \mathcal{H}_Y \neq 0$. In particular:

$$\{X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})\colon \mathcal{H}_X \neq 0\} \neq \emptyset.$$

By well-foundedness of $\mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ (Corollary 2.6), we can fix a minimal element X in this set.

Claim. *For every $g, h \in G$ such that $gG_{\{X\}} \neq hG_{\{X\}}$, we have:*

$$\pi(g)\mathcal{H}_X \perp \pi(h)\mathcal{H}_X\,. \tag{2}$$

Proof of the claim. Suppose first that $X \in \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$. Let x be the unique element of $X \cap \mathbf{T}$. Then, $X = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)_x$ hence $g(X) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)_{g(x)}$ and $h(X) = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)_{h(x)}$ with $g(x) \neq h(x)$. In this case, the claim follows directly from Lemma 2.9 (recall that π is non-spherical). We can thus assume that $X \notin \mathcal{C}_0(\mathbf{T}^{\mathrm{eq}})$. Let $\xi, \eta \in \mathcal{H}_X$. Then:

$$\langle \pi(g)\xi, \pi(h)\eta\rangle = \left\langle \pi(h^{-1}g)p_X\xi, p_X\eta\right\rangle = \left\langle p_{h^{-1}g(X)}\pi(h^{-1}g)\xi, p_X\eta\right\rangle = \left\langle \pi(h^{-1}g)\xi, p_{h^{-1}g(X)}p_X\eta\right\rangle. \tag{3}$$

Recalling that X is locally finite and $h^{-1}g \notin G_{\{X\}}$, we get $X \not\subseteq h^{-1}g(X)$. Applying Lemma 2.11 and by minimality of X, we get $p_{h^{-1}g(X)}p_X = 0$. Equation (3) now yields $\langle \pi(g)\xi, \pi(h)\eta\rangle = 0$ and the claim follows.

$\square_{\text{claim}}$

Finally, let $X_0 \subseteq \mathbf{T}^{\mathrm{eq}}$ be a finite subtree such that $\mathcal{H}_{X_0} \neq 0$. Up to replacing X_0 with a larger set, Lemma 2.4 allows us to assume that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X$ and that X_0 is $G_{\{X\}}$-stable. Now, $G_{\{X\}}$ acts on $\mathcal{H}_{X_0}$, an action which factorizes into a representation σ of the finite group $G_{\{X\}}/G_{X_0}$. Form $\pi' := \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)$ with underlying space $\mathcal{H}'$. Let $(g_i)_{i\in I}$ be a system of coset representatives of $G/G_{\{X\}}$. Define a map $\Phi\colon \mathcal{H}' \longrightarrow \mathcal{H}$ in the following way:

$$\forall f \in \mathcal{H}',\ \Phi(f) = \sum_{i\in I} \pi(g_i)f(g_i)\,.$$

By definition of the norm on $\mathcal{H}'$ and by Equation (2), Φ is well defined and isometric. Note that it does not depend on the choice of (g_i), for if $h^{-1}g \in G_{\{X\}}$, then $\pi(g)f(g) = \pi(g)f(hh^{-1}g) = \pi(g)(\sigma(g^{-1}h)f(h)) = \pi(g)\pi(g^{-1}h)f(h) = \pi(h)f(h)$. With similar calculations, it is easily seen that Φ is G-equivariant hence an isomorphism of representations.

To conclude, recall that $G_{\{X\}}/G_{X_0}$ is a finite group hence σ splits as a sum of irreducible subrepresentations. Moreover, by the basic properties of induction, if $\tau \subseteq \sigma$, then $\mathrm{Ind}_{G_{\{X\}}}^{G}(\tau) \subseteq \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)$ hence σ can indeed be assumed irreducible. $\square$

We are now ready to state and prove our main theorem. It readily implies Theorem 1 from the Introduction.

Theorem 2.13. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive.*

1. *Let X_0 be a non-empty finite subtree of $\mathbf{T}^{\mathrm{eq}}$ and let $X := \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0)$. Let $\sigma\colon G_{\{X\}} \curvearrowright \mathcal{K}$ be an irreducible unitary representation of $G_{\{X\}}$ that is trivial on G_{X_0}. Then $\pi := \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)$ is an irreducible representation of G.*
2. *A representation of the above form is spherical if and only if either*

- *X is empty, in which case π is the trivial representation of G,*
- *or $G_{\{X\}} = G_x$ for some $x \in \mathbf{T}$ and σ is the trivial representation of G_x, in which case π is isomorphic to $G \curvearrowright \ell^2(G/G_x)$.*

3. *Every non-spherical irreducible representation of G is of the above form.*
4. *Two representations of the above form $\mathrm{Ind}_{G_{\{X\}}}^G(\sigma)$ and $\mathrm{Ind}_{G_{\{Y\}}}^G(\tau)$ are isomorphic if and only if there exists $g \in G$ such that $g(X) = Y$ and $\tau^g \simeq \sigma$.*
5. *Every non-spherical representation of G splits as a sum of non-spherical irreducible representations of the above form.*

Proof. Items 1, 2 and 4 correspond to Proposition 2.8. Item 5 is a straightforward combination of Corollary 2.12 and Zorn's Lemma. It only remains to prove Item 3. To that aim, let π be a non-spherical irreducible unitary representation of G. By Corollary 2.12, there exists a π' of the desired form such that $\pi' \subseteq \pi$. Since π is irreducible, we have $\pi' \simeq \pi$. □

It is worth pointing out that Corollary 2.12 combined with Zorn's Lemma also has the following consequence. It can be interpreted as a form of dissociation for the representations of G. It is the exact analogue of Lemma 1.22 and weaker than dissociation as defined in [BJJ24].

Corollary 2.14. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical structure and 2-transitive. Let $\pi\colon G \curvearrowright \mathcal{H}$ be a non-spherical representation of G. For every algebraically closed subsets $X, Y \subseteq \mathbf{T}^{\mathrm{eq}}$, we have*

$$\mathcal{H}_X \perp_{\mathcal{H}_{X\cap Y}} \mathcal{H}_X.$$

An important consequence of the results in this section is the fact that G is of Type I when $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. Before proving this fact, we formalize this notions. See e.g. [Mac76] or [Bd20, Chap.6] for a complete exposition, the beginning of which does not assume local compactness.

Let H be a topological group and let $\pi\colon H \curvearrowright \mathcal{H}$ and $\pi'\colon H \curvearrowright \mathcal{H}'$ be two unitary representations of H. We will say that π and π' are *disjoint* if no non-zero subrepresentation of π is isomorphic to a subrepresentation of π'. If π has the property that, whenever $\pi \simeq \pi_1 \oplus \pi_2$, the representations π_1 and π_2 are always disjoint, we will say that π is *multiplicity free*. For instance, every irreducible representation is multiplicity free and more generally so is every sum of pairwise non-isomorphic irreducible representations.

The representations π and π' are *quasi-equivalent* if they are isomorphic up to multiplicity, that is if $\oplus_{n\in\mathbb{N}}\pi \simeq \oplus_{n\in\mathbb{N}}\cdot\pi'$. Then π is of *Type I* if π is quasi-equivalent to a multiplicity free representation. This is equivalent to the von Neumann algebra $\pi(G)' := \{T \in \mathcal{B}(\mathcal{H})\colon \forall g \in G,\ T\pi(g) = \pi(g)T\}$ being of Type I in the usual terminology [Dix96, Prop. 5.4.7]. Moreover, if π is of Type I, the multiplicity free representation to which it is quasi equivalent is unique up to isomorphism and is a subrepresentation of π. Finally, the group H is of *Type I* if all of its representations are of Type I.

The proof of the following result follows closely Ol'šanskiĭ's proof in the special case $G = \mathrm{Aut}(\mathbf{T})$ [Ol'80, Th. 5.7].

Corollary 2.15. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive.*
Then G is of Type I.

Proof. Let $\pi\colon G \curvearrowright \mathcal{H}$ be a representation of G. Let $\mathcal{K}$ be the closed subspace of $\mathcal{H}$ generated by $\bigcup_{x\in\mathbf{T}} \mathcal{H}_x$. Let π_1 and π_2 be the restrictions of π to $\mathcal{K}$ and $\mathcal{K}^\perp$ respectively. By [Bd20, Prop. 6.B.13(2)], sums of Type I representations are of Type I. It thus suffices to check that both π_1 and π_2 are of Type I. We will thus assume that either $\pi = \pi_1$ of $\pi = \pi_2$.

We start with the case $\pi = \pi_2$. This representation is non-spherical hence splits as a sum of irreducible subrepresentations by Corollary 2.14. Irreducible representations being multiplicity free, they are of Type I. We conclude with [Bd20, Prop. 6.B.13(2)].

We now assume that $\pi = \pi_1$ and show that $\pi(G)'$ is a Type I von Neumann algebra. Fixing $x \in \mathbf{T}$, let $\mathcal{A}_x$ be the von Neumann algebra generated by $p_x\pi(G)p_x$. The operator p_x induces an isomorphism between $\pi(G)'$ and the commutant of $\mathcal{A}_x$. It thus suffices to show that the commutant of $\mathcal{A}_x$ is of Type I which in turn reduces to showing that $\mathcal{A}_x$ itself is of Type I [Tak02,

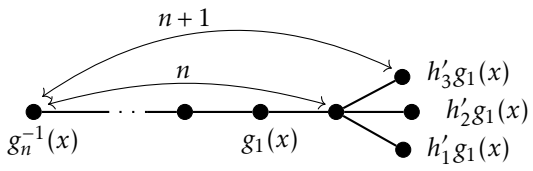

FIGURE 4. Computing the distances $d(h_i'g_1(x), g_n^{-1}(x))$.

Cor. 1.30]. In fact, we will show that $\mathcal{A}_x$ is commutative, which is enough to conclude [Tak02, Prop. 1.23].

We now follow the proof of [Ol'80, Sec 5.3], using the mentioned alterative method due to Ismagilov, to show that $\mathcal{A}_x$ is commutative. For every $n \in \mathbb{N}$, let $g_n \in G$ be such that $d(g_n(x), x) = n$. Such a sequence can be obtained for instance by transitivity of $G \curvearrowright \mathbf{T}$ (Item 1 of Lemma 1.7). As seen in the proof of Corollary 1.13 (first argument in $(iv) \Rightarrow (ii)$), the double coset $G_x g G_x$ for $g \in G$ only depends on $d(g(x), x)$. In particular, $a_n := p_x \pi(g_n) p_x$ does not depend on the choice of g_n, and $\{a_n \colon n \in \mathbb{N}\} = p_x \pi(G) p_x$. Moreover, since $d(g(x), x) = d(x, g^{-1}(x))$ for every $g \in G$, we have $a_n^* = a_n$ for every $n \in \mathbb{N}$. Thus, the linear span of $\{a_n \colon n \in \mathbb{N}\}$ is dense in $\mathcal{A}_x$ and it suffices to show that the a_n's pairwise commute. It is a direct consequence of the following claim:

Claim. *For every $n \in \mathbb{N}$, $a_{n+1} = a_n a_1$.*

Proof of the claim. Let $n \in \mathbb{N}$. Up to replacing g_1, we can assume that $g_1(x) \in [x, g_n^{-1}(x)]$. Let $\xi \in \mathcal{H}$ and let $\varepsilon > 0$. The Alaoglu-Birkhoff Theorem 1.18 applied to $\eta := \pi(g_1) p_x \xi$, yields $\lambda_1, \dots, \lambda_m \geqslant 0$ with $\lambda_1 + \cdots + \lambda_m = 1$ and $h_1, \dots, h_m \in G_x$ such that:

$$\|p_x \eta - \sum_{i=1}^{m} \lambda_i \pi(h_i) \eta\| < \varepsilon. \tag{4}$$

By edge transitivity of $G \curvearrowright \mathbf{T}$ (Item 2 in Lemma 1.7), pick $h \in G_x$ such that $hg_1(x) \notin \{h_1 g_1(x), \dots, h_m g_1(x)\}$ and let $h_i' := h^{-1} h_i$ for every $i \leqslant m$. It satisfies $d(g_n h_i' g_1(x), x) = d(h^{-1} h_i g_1(x), g_n^{-1}(x)) = n+1$, see Figure 4. In particular, $p_x \pi(g_n h_i' g_1) p_x = a_{n+1}$ for every $i \leqslant m$. Moreover, since $h \in G_x$, replacing h_i with h_i' for every $i \leqslant m$ preserves (4). Thus:

$$\begin{aligned}
\|a_{n+1}\xi - a_n a_1 \xi\| &= \|a_{n+1}\xi - p_x \pi(g_n) p_x \eta\| \\
&< \varepsilon + \|a_{n+1}\xi - p_x \pi(g_n) \cdot \sum_{i=1}^{m} \lambda_i \pi(h_i') \eta\| \\
&< \varepsilon + \sum_{i=1}^{m} \lambda_i \|a_{n+1}\xi - p_x \pi(g_n h_i' g_1) p_x \xi\| \\
&< \varepsilon + \sum_{i=1}^{m} \lambda_i \|a_{n+1}\xi - a_{n+1}\xi\| \\
&< \varepsilon\,.
\end{aligned}$$

Since ε is arbitrary, the claim follows. $\square_{\text{claim}}$

$\square$

3. The Howe-Moore property

This Section is fully independent from Section 2 and does not use the expansion of $\mathbf{T}$ into $\mathbf{T}^{\text{eq}}$, with the exception of our very last results, Lemma 3.8 and Corollary 3.9. Its goal is to show that the representation theory of G interacts well with its large scale geometry. In Subsection 3.1, we extend the definition of the Howe-Moore property to general topological groups with a special focus on locally Roelcke-precompact Polish groups. In Subsection 3.2, we recall classical results about the geometry of $\mathbf{T}$ and provide a proof of Theorem 2 from the introduction, namely that

G^+ has the Howe-Moore property. We also show that this result combined with those of Section 2 provides a deep understanding of the lattice of open subgroups of G.

3.1. **The Howe-Moore property for locally Roelcke-precompact Polish groups.**

Recall that given a representation $\pi\colon K \curvearrowright \mathcal{H}$ of a topological group K, an *invariant vector* is a fixed point of the action. Moreover, the *matrix coefficients* of K arising from π are the maps $K \longrightarrow \mathbb{C}$ of the form $f^{\pi}_{\xi,\eta}\colon k \longmapsto \langle \pi(k)\xi, \eta\rangle$ for every $\xi, \eta \in \mathcal{H}$. These continuous maps capture the whole representation theory of K [Bd20, Sec. 1.B] and we will show that they also preserve the large scale geometry of universal Burger-Mozes groups. Initially defined for locally compact groups, the Howe-Moore property captures precisely that. While it naturally extends to general topological groups, it has not yet been studied in a non-locally compact setting before to the best of the author's knowledge, except in [BJJ25].

Definition 3.1. Let K be a topological group.

1. A continuous map $f : K \longrightarrow \mathbb{C}$ *vanishes at infinity* if for every $\varepsilon > 0$, the set $\{k \in K\colon |f(k)| \leqslant \varepsilon\}$ is coarsely bounded. We will denote by $C_0(K)$ the set of such functions.
2. The group K has the *Howe-Moore property* if for every representation $\pi\colon K \curvearrowright \mathcal{H}$ with no non-zero invariant vector, all the matrix coefficients of K arising from π vanish at infinity.

In fact, this notion can often be rephrased using more familiar objects, namely locally compact ones. Given a topological group K, its *Roelcke completion* $\mathcal{R}(K)$ is the completion of K with respect to the meet of the left and right uniformities on K. Another way to describe this object when K is metrizable is as follows. Let δ_L be any left-invariant metric on K that generates its topology and set for every $g, h \in K$:

$$\delta_{L\wedge R}(g,h) = \inf\{\delta_L(g,k) + \delta_L(k^{-1},h^{-1})\colon k \in K\}.$$

The metric $\delta_{L\wedge R}$ is no longer left-invariant but it still generates the topology of K. Moreover, $\mathcal{R}(K)$ is the completion of the metric space $(K, \delta_{L\wedge R})$. It is straightforward that a Polish group K is Roelcke precompact if and only if $\mathcal{R}(K)$ is compact. A deeper result due to Zielinski is that K is locally Roelcke-precompact if and only if $\mathcal{R}(K)$ is locally compact [Zie21]. In this case, we define $C_0(\mathcal{R}(K))$ in the classical way, as the set of continuous functions $f\colon \mathcal{R}(K) \longrightarrow \mathbb{C}$ such that for every $\varepsilon > 0$, the set $\{x \in \mathcal{R}(K)\colon |f(x)| \geqslant \varepsilon\}$ is compact.

Back to unitary representations, it easily checked that every matrix coefficient of a metrizable group K is *Roelcke-uniformly continuous*, i.e. uniformly continuous with respect to $\delta_{L\wedge R}$. In particular, every matrix coefficients $K \longrightarrow \mathbb{C}$ uniquely extends to a continuous function $\mathcal{R}(K) \longrightarrow \mathbb{C}$.

Proposition 3.2. *Let K be a locally Roelcke-precompact Polish group. The following properties are equivalent:*

(i) K has the Howe-Moore property,
(ii) for every unitary representation $\pi\colon G \curvearrowright \mathcal{H}$ with no non-zero invariant vector, all the matrix coefficients of K arising from π vanish at infinity on the locally compact space $\mathcal{R}(K)$.

Given a sequence $(g_n)_{n\in\mathbb{N}}$ in a topological group K, we will write $g_n \longrightarrow \infty$ if $(g_n)_{n\in\mathbb{N}}$ eventually escapes every coarsely bounded subset of K. The large scale geometry of our groups is nice enough so that it is fully captured by sequences.

Lemma 3.3. *Assume $\mathfrak{M}$ is $\aleph_0$ categorical and 2-transitive. Let $f \in C(G^+)$. The following properties are equivalent:*

(i) f vanishes at infinity,
(ii) for every $(g_n) \in {G^+}^{\mathbb{N}}$, $g_n \longrightarrow \infty$ implies $f(g_n) \longrightarrow 0$.

Proof. $(i) \Rightarrow (ii)$ is obviously true in higher generality. Suppose now that f does not satisfy (i). Then, there exists $\varepsilon > 0$ such that $N := \{g \in G^+\colon |f(g)| \geqslant \varepsilon\}$ is not coarsely bounded. By Corollary 1.13, $N \cdot x$ is unbounded in $\mathbf{T}$. Thus, for every $n \in \mathbb{N}$, there exists $g_n \in N$ such that $d(g_n(x), x) \geqslant n$. We have $g_n \longrightarrow \infty$ by Corollary 1.13 but $|f(g_n)| \geqslant \varepsilon$, contradicting (ii). □

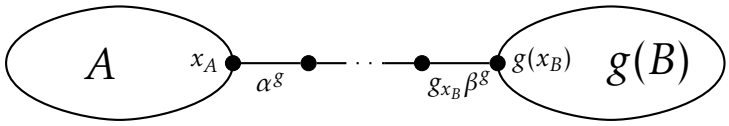

FIGURE 5. The local data used for identifying $G_A g G_B$.

3.2. **Ends, translations, and the Howe-Moore property for G^+.**

We now recall some terminology and classical facts regarding the geometry of $\mathbf{T}$ itself. To that aim, we start by defining the boundary $\partial\mathbf{T}$ of $\mathbf{T}$. Given two infinite injective paths $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}} \in \mathbf{T}^{\mathbb{N}}$, we will write $(x_n)_{n\in\mathbb{N}} \simeq (y_n)_{n\in\mathbb{N}}$ if they eventually coincide up to a shift of the indices. That is, if there exists $k \in \mathbb{Z}$ and $N \in \mathbb{N}$ such that $x_{n+k} = y_n$ for every $n \geqslant N$. Then $\partial\mathbf{T}$ will denote the set of $\simeq$-equivalence classes of infinite injective paths. The set $\partial\mathbf{T}$ will be called the *boundary* of $\mathbf{T}$ and its elements the *ends* of $\mathbf{T}$. Given $\omega, \omega' \in \partial\mathbf{T}$, there exists a bi-infinite injective path $(x_n)_{n\in\mathbb{Z}}$ such that $(x_n)_{n\in\mathbb{N}} \in \omega'$ and $(x_{-n})_{n\in\mathbb{N}} \in \omega$. Moreover, this sequence is unique up to shift of the indices hence we will denote by $]\omega, \omega'[$ the set of vertices appearing in such a sequence.

A *translation* of $\mathbf{T}$ is a tree automorphism that neither fixes a vertex nor flips an edge. Given a translation $\tau \in \mathrm{Aut}(\mathbf{T})$ and $x \in \mathbf{T}$ such that $d(\tau(x), x)$ is minimal, let $\Delta := \bigcup_{n\in\mathbb{Z}} \tau^n([x, \tau(x)])$. Then, Δ is stable under the action of τ and there exists $\omega \neq \omega' \in \partial\mathbf{T}$ such that $\Delta =]\omega, \omega'[$. The set Δ is called the *axis* of τ and $n := d(\tau(x), x)$ is the *step* of τ. Note that $\tau \in \mathrm{Aut}(\mathbf{T})^+$ if and only if it is of even step. See e.g. [FN91, Chap. 1] for more details. Applying Corollary 1.4 to $X =]\omega, \omega'[$ for any $\omega \neq \omega' \in \partial\mathbf{T}$, it is easy to see that G^+ does contain translations of step 2 when $\mathfrak{M}$ is 2-transitive.

Our proof that G^+ has the Howe-Moore property is adapted from the classical one in the finite arity case. The main difference is that since our groups are not locally compact, a finer Cartan decomposition is required. More precisely, we give a description of $G_A \backslash G^+ / G_B$ for every finite subtrees $A, B \subseteq \mathbf{T}$, while the case $A = B = \{x\}$ suffices in finite arity. When $A = B = \{x\}$, the graph distance is enough to describe $G_x \backslash G^+ / G_x$. In the general case, additional information is required, which we now describe.

From now on, we fix two non-empty finite subtrees A and B of $\mathbf{T}$. Let $n := |A|$, $m := |B|$ and write $A = \{a_1, \ldots, a_n\}$, $\overline{a} := (a_1, \ldots, a_n)$, $B = \{b_1, \ldots, b_m\}$ and $\overline{b} := (b_1, \ldots, b_m)$. The symbol $\wedge$ will denote concatenation of tuples. For $a \in A$, we will write $A(a)$ instead of $E_A(a)$ and similarly for B. Given $g \in G^+$ such that $g(B) \cap A = \emptyset$, let x_A^g and x_B^g be the unique vertices of A and B respectively such that $d(A, g(B)) = d\left(x_A^g, g\left(x_B^g\right)\right)$. Let also α^g be the element of $\mathfrak{M}$ coloring the unique edge going from x_A^g toward $g(B)$. Similarly, let β^g be the element of $\mathfrak{M}$ coloring the unique edge around x_B whose image under g goes toward A. Write $t_A^g := \Gamma_{A(x_A^g)} \cdot \alpha^g$ and $t_B^g := \Gamma_{B(x_B^g)} \cdot \beta^g$. See Figure 5. Finally, define:

$$\Psi(g) := \left(x_A^g, x_B^g, t_A^g, t_B^g, d\left(A, g(B)\right)\right).$$

Lemma 3.4. *Assume $\mathfrak{M}$ is 2-transitive. Let $g, h \in G^+$ be such that $g(B) \cap A = \emptyset$ and $h(B) \cap A = \emptyset$. The following properties are equivalent:*

(i) $G_A g G_B = G_A h G_B$,
(ii) $\Psi(g) = \Psi(h)$.

Proof. Clearly, Ψ is left G_A-invariant and right G_B-invariant hence $(i) \Rightarrow (ii)$. Conversely, suppose $\Psi(g) = \Psi(h)$. We will use the following simplified notations: $x_A := x_A^g = x_A^h$ and $x_B := x_B^g = x_B^h$. Let $(x_0, \ldots, x_{d+1})$ be the shortest path from A to $g(B)$ and let $(y_0, \ldots, y_{d+1})$ be the shortest path from A to $h(B)$. Necessarily, we have $x_0 = y_0 = x_A$ as well as $x_{d+1} = g(x_B)$ and $y_{d+1} = h(x_B)$. Let C be the finite subtree of $\mathbf{T}$ generated by A and $g(B)$, i.e. $C = A \cup g(B) \cup \{x_1, \ldots, x_d\}$. We define $(\gamma_c)_{c\in C}$ be as follows (see also Figure 6):

– Since $t_A^g = t_A^h$, there exists $\gamma_{x_A} \in \Gamma_{A(x_A)}$ such that $\gamma_{x_A}(\alpha^g) = \alpha^h$. For every $a \in A \backslash \{x_A\}$, set $\gamma_a := \mathrm{id}_{\mathfrak{M}}$.

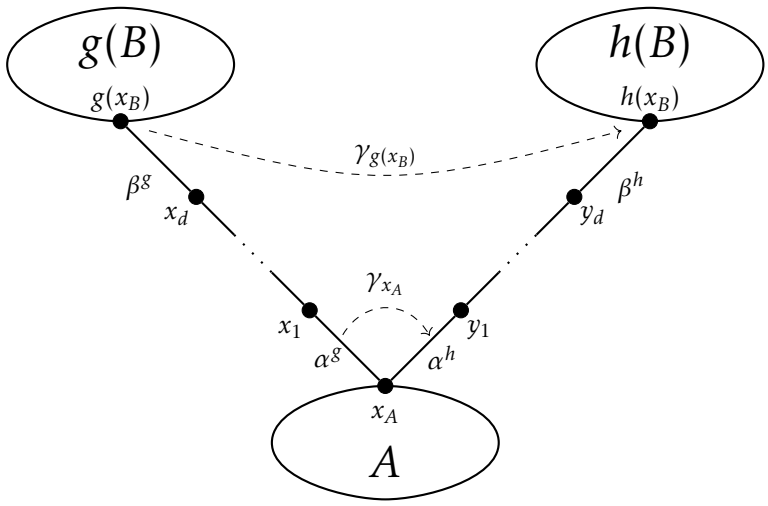

FIGURE 6. Local actions at x_A and $g(x_B)$. At the bottom, γ_{x_A} fixes $A(x_A)$ and sends α^g to α^h. Up above, $\gamma_{g(x_B)}$ reverses the action of g_{x_B} on $B(x_B)$ and replaces it with that of h_{x_B} while sending β^g to β^h.

- Similarly, since $t_B^g = t_B^h$, there exists $\gamma \in \Gamma_{B(x_B)}$ such that $\gamma \cdot \beta^g = \beta^h$. Set $\gamma_{g(x_B)} := h_{x_B}\gamma\left(g_{x_B}\right)^{-1}$ and, for every $b \in B\backslash\{x_B\}$, set $\gamma_{g(b)} := h_b\,(g_b)^{-1}$.
- By 2-transitivity of $\mathfrak{M}$, we can pick for every $i \in \{1,\dots,d\}$ some $\gamma_{x_i} \in \Gamma$ such that $\gamma_{x_i} \cdot \chi(x_i, x_{i+1}) = \chi(y_i, y_{i+1})$ and $\gamma_{x_i} \cdot \chi(x_i, x_{i-1}) = \chi(y_i, y_{i-1})$.

Then $(\gamma_c)_{c\in C}$ satisfies the conditions of Corollary 1.4 which yields $u \in G$ such that $u_c = \gamma_c$ for every $c \in C$ and $u(x_A) = x_A$. By construction, $u \in G_A$ and $ug(b) = h(b)$ for every $b \in B$. Thus $(ug)^{-1}h \in G_B$ and $h \in ugG_B \subseteq G_AgG_B$ hence $G_AgG_B = G_AhG_B$.

□

Note that if we assume that $\mathfrak{M}$ is $\aleph_0$-categorical, then $\Gamma \curvearrowright \mathfrak{M}$ is oligomorphic and the first four components of Ψ take values in a finite set. From now on, we fix $\tau \in G^+$ a translation of step 2, which will allow us to control the last component of Ψ.

Lemma 3.5. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive. There exists a finite subset $F \subseteq G^+$ such that:*

$$G^+ = G_AF\{\tau^n : n \geqslant 0\}FG_B.$$

Proof. Let $K := \{g \in G^+ : g(B) \cap A \neq \emptyset\}$. Then K is Roelcke precompact in G^+ by Lemma 1.12. Thus, there exists a finite subset $F_0 \subseteq K \subseteq G^+$ such that $K \subseteq G_AF_0G_B$. It remains to deal with $G^+\backslash K$.

With an eye towards Lemma 3.4, fix $a \in A$, $b \in B$ as well as $t_A = \Gamma_{A(a)} \cdot \alpha$ and $t_B = \Gamma_{B(b)} \cdot \beta$ for some $\alpha \in \mathfrak{M}\backslash A(a)$ and $\beta \in \mathfrak{M}\backslash B(b)$ respectively. Let also $\delta = 0$ if $d(a,b)$ is even or $\delta = 1$ if $d(a,b)$ is odd. We will produce $g = g(b, t_B)$ and $h = h(a, t_A)$ in G^+ such that for every $n \in \mathbb{N}$,

$$\Psi(h\tau^n g) = (a, b, t_A, t_B, 2(n+1) - \delta).$$

Note that for every $g \in G^+$, we have $d(a, g(b)) \mod 2 = d(a,b) \mod 2$ hence this will be enough to conclude, after setting:

$$F := F_0 \cup \bigcup_{a,b,t_A,t_B} \{g(b,t_B), h(a,t_A)\}.$$

Since the action $\Gamma \curvearrowright \mathfrak{M}$ is oligomorphic, this is indeed a finite subset of G^+.

Let c be the unique neighbor of b such that $\chi(b,c) = \beta$. Let $\Delta := \{x_i : i \in \mathbb{Z}\}$ be the axis of τ, written in such a way that $\tau(x_i) = x_{i+2}$ for every $i \in \mathbb{Z}$ and $d(x_0, b)$ is even. By Item 3 of Lemma

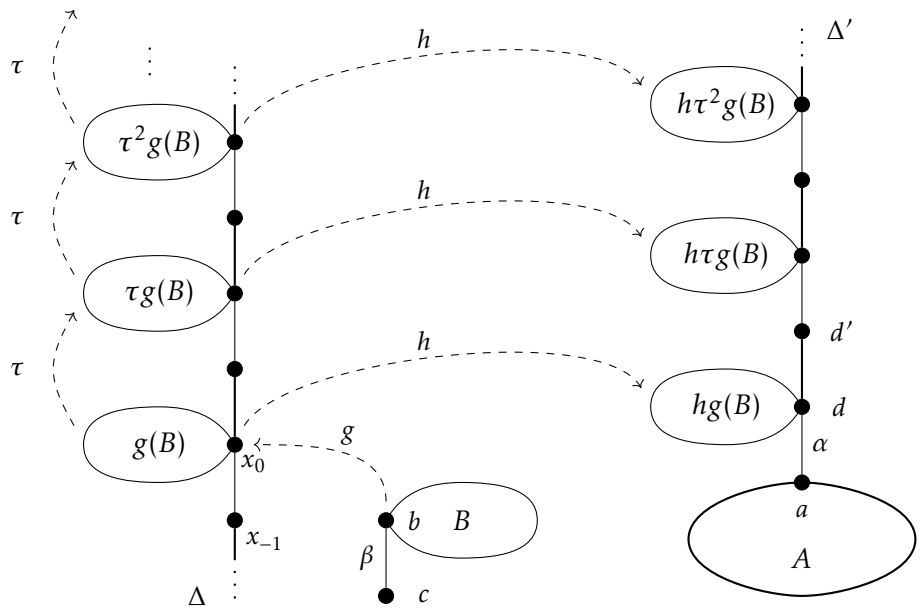

FIGURE 7. Realization of a given value of Ψ.

1.7, $G \curvearrowright \mathbf{T}$ is 2-transitive and we can find $g \in G$ such that $g(b) = x_0$ and $g(c) = x_{-1}$. By choice of x_0 we automatically have $g \in G^+$.

Next, let d be the unique neighbor of a such that $\chi(a,d) = \alpha$, let d' be any neighbor of d distinct from a. Pick Δ' a bi-infinite geodesic in $\mathbf{T}$ such that $\Delta' \cap A = \{a\}$ and $d, d' \in \Delta$. Using 2-transitivity of $\mathfrak{M}$ and Corollary 1.4, one can find $h \in G$ such that:

- $h(\Delta) = \Delta'$,
- $(h(x_{-1}), h(x_0)) = \begin{cases} (a,d) & \text{if } d(x_0,d) \text{ is even,} \\ (d,d') & \text{otherwise.} \end{cases}$

Again, by parity of $d(h(x_0), x_0)$, we in fact have $h \in G^+$. The case where $d(x_0,d)$ is even is represented in Figure 7. Since for every $n \in \mathbb{N}$, no edge of $h\tau^n g(B)$ ever belongs to Δ', we indeed have:

$$\forall n \in \mathbb{N},\ \Psi(h\tau^n g) = (x_A, x_B, t_A, t_B, 2(n+1) - \delta)\,.$$

□

Just as in the finite arity case, the previous Lemma reduces the problem to studying the vanishing of matrix coefficients of G on the abelian semigroup $\{\tau^n \colon n \in \mathbb{N}\}$.

Corollary 3.6. *Let $\pi \colon G^+ \curvearrowright \mathcal{H}$ be a representation of G^+ and suppose there exists $\xi, \eta \in \mathcal{H}$ such that $f^\pi_{\xi,\eta}$ does not vanish at infinity. Then, there exists $\xi', \eta' \in \mathcal{H}$ such that:*

$$f^\pi_{\xi',\eta'}(\tau^n) \underset{n\to+\infty}{\not\longrightarrow} 0\,.$$

Proof. Suppose $f^\pi_{\xi,\eta}$ does not vanish at infinity. Approximating ξ and η by invariant vectors using Lemma 1.21 yields a uniform approximation of f. We can thus assume $\xi, \eta \in \mathcal{H}_A$ for some non-empty finite subtree $A \subseteq \mathbf{T}$. By Lemma 3.3, there exists $\varepsilon > 0$ and $(g_n) \in G^{+\mathbb{N}}$ such that $g_n \longrightarrow \infty$ while $\left|f^\pi_{\xi,\eta}(g_n)\right| \geqslant \varepsilon$ for every $n \in \mathbb{N}$. By Lemma 3.5, we have $G^+ = G_A F\{\tau^n \colon n \in \mathbb{N}\} F G_A$ for some finite subset $F \subseteq G^+$ and we can write $g_n = u_n f_n \tau^{k_n} f'_n u'_n$ for some $u_n, u'_n \in G_A$, $f_n, f'_n \in F$ and $k_n \in \mathbb{N}$. Up to extraction, we can assume that (f_n) and (f'_n) are constant sequences and denote their respective value by f and f'. Observe using Corollary 1.13 that G_A and F are coarsely bounded and that $k_n \longrightarrow +\infty$.

Setting $\xi' := \pi(f')\xi$ and $\eta' := \pi(f^{-1})\eta$ we have for every $n \in \mathbb{N}$:

$$f^{\pi}_{\xi,\eta}(g_n) = f(u_n f \tau^{k_n} f' u'_n) = \left\langle \pi(\tau^{k_n})\pi(f')\pi(u'_n)\xi, \pi(f^{-1})\pi(u_n^{-1})\eta \right\rangle = \left\langle \pi(\tau^{k_n})\xi', \eta' \right\rangle = f^{\pi}_{\xi',\eta'}(\tau^{k_n}).$$

We thus have $\left| f^{\pi}_{\xi',\eta'}(\tau^{k_n}) \right| \geqslant \varepsilon$ for every $n \in \mathbb{N}$ while $k_n \longrightarrow +\infty$. In particular, $f^{\pi}_{\xi',\eta'}(\tau^n) \not\longrightarrow 0$. □

The classical proof of Lubotzky and Mozes now carries through verbatim. More precisely, Propositions 1 and 2 in [LM92] hold for G^+. Once combined, they yield the following, which completes the proof of Theorem 2:

Proposition 3.7. *Let $\pi\colon G^+ \curvearrowright \mathcal{H}$ be a representation of G^+ with no non-zero G^+-invariant vector. Then $f^{\pi}_{\xi,\eta}(\tau^n) \longrightarrow 0$ for every $\xi, \eta \in \mathcal{H}$.*

Proof. See Propositions 1 and 2 in [LM92]. □

3.3. **Commensurators and open subgroups of G.**

Combining the results of this section with those of Section 2, we can describe the open subgroups of G, at least up to finite index. In model theoteric terms, the results below imply that every imaginary of $\mathbf{T}$ is inter-algebraic with some finite subtree of $\mathbf{T}^{\mathrm{eq}}$.

In what a follows, given two subgroups $H, H' \leqslant G$, we will say that H and H' are *commensurate* if $H \cap H'$ has finite index in both H and H', which defines an equivalence relation. The *commensurator* of H is the group $\mathrm{Comm}_G(H) := \{g \in G\colon H \text{ and } gHg^{-1} \text{ are commensurate}\}$. For every $g \in G$, the number of left H-cosets in HgH is precisely the index $[H\colon H \cap gHg^{-1}]$ while the number of right cosets is $[gHg^{-1}\colon H \cap gHg^{-1}]$. If H and H' are commensurate, it is easily seen that $\mathrm{Comm}_G(H) = \mathrm{Comm}_G(H')$. We will say that H is *autocommensurate* in G if $H = \mathrm{Comm}_G(H)$.

Lemma 3.8. *Suppose $\mathfrak{M}$ is 2-transitive and $\aleph_0$-categorical. Let $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$. Then $G_{\{X\}}$ is autocommensurate.*

Proof. Let $g \in \mathrm{Comm}_G(G_{\{X\}})$. Let $X_0 \subseteq X$ be finite and such that $\mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X$ (Lemma 2.4). Since G_{X_0} has finite index in $G_{\{X\}}$, we also have $g \in \mathrm{Comm}_G(G_{X_0})$. In particular, the number of left-G_{X_0} cosets contained in $G_{X_0} g G_{X_0}$ is finite and $G_{X_0} \cdot g(X_0)$ is finite. Equivalently, $g(X_0) \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(X_0) = X$ hence $g(X) \subseteq X$. Since X is locally finite, we in fact have $g(X) = X$, i.e. $g \in G_{\{X\}}$.

Conversely, suppose $g \in G_{\{X\}}$. Since G_{X_0} has finite index in $G_{\{X\}}$, it suffice to show that $g \in \mathrm{Comm}_G(G_{X_0})$. Write $X_0 = \{x_1, \ldots, x_n\}$, $\overline{x} := (x_1, \ldots, x_n)$ and simply observe that the number of left G_{X_0}-cosets in $G_{X_0} g G_{X_0}$ is the cardinal of the orbit $G_{X_0} \cdot g(\overline{x})$. Since $g(X_0) \subseteq X$, this orbit is finite. Similarly, the number of right G_{X_0}-cosets in $G_{X_0} g G_{X_0}$ is the cardinal of the orbit $G_{X_0} \cdot g^{-1}(\overline{x})$ which is also finite. □

Corollary 3.9. *Assume $\mathfrak{M}$ is $\aleph_0$-categorical and 2-transitive and let $U \leqslant G$ be a proper open subgroup. Then, either $U = G^+$ or U is Roelcke precompact. In the latter case, there exists $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ such that $U \leqslant G_{\{X\}}$ with finite index.*

Proof. If $G^+ \subseteq U$ then $G^+ = U$ because G^+ has index 2 in G. We thus suppose that $G^+ \not\subseteq U$. We first show that U is Roelcke precompact. Since $U \cap G^+$ has index at most 2 in U, it suffices to prove that $U \cap G^+$ is Roelcke precompact. It is a classical consequence of the Howe-Moore property that every proper open subgroup of G^+ is coarsely bounded, see e.g. [BJJ25, Lem. 2.9] for a proof. In particular, $U \cap G^+$ is coarsely bounded hence Roelcke precompact by [Zie21, Th. 14].

As for the second statement, we will first show that $H := \mathrm{Comm}_G(U)$ is Roelcke precompact. Since U is open, there exists a finite subtree $Z \subseteq \mathbf{T}$ such that $G_Z \leqslant U$. Since $\mathrm{acl}_{\mathbf{T}}(Z)$ is finite (apply e.g. Proposition 2.5 and Corollary 2.3), we can assume that Z is algebraically closed with respect to $G \curvearrowright \mathbf{T}$. Next, Item (v) of Corollary 1.13 provides a subtree $Y \subseteq \mathbf{T}$ which is either a single vertex or a single edge and such that $U \leqslant G_{\{Y\}}$. For every $g \in H$, the number of distinct left-cosets of U contained in UgU is $[U, U \cap U^g]$ hence finite. In particular, $UgU \cdot Y$ is finite and, a fortiori, so is $G_Z \cdot g(Y)$. Thus $g(Y) \subseteq Z$ and $H \cdot Y \subseteq Z$ is bounded. In particular, H is indeed Roelcke precompact by Corollary 1.13. Now, $\mathrm{Comm}_G(U) = \mathrm{Comm}_H(U)$ and [Tsa12, Lem. 2.7] applies, hence U has finite index in H and H is autocommensurate. It remains to show that $H = G_{\{X\}}$ for some $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$.

It is easily seen that $\lambda_{G/H}\colon G \curvearrowright \ell^2(G/H)$ is irreducible. We distinguish two cases.

- Suppose that $\lambda_{G/H}$ is spherical. Let $x \in \mathbf{T}$ and $\xi \in \mathcal{H}_x\backslash\{0\}$. There exists $g \in G$ such that $\xi(gH) \neq 0$. Since $\xi \in \ell^2(G/H)$, $G_x gH$ contains only finitely many left cosets of H. Then $x' := g^{-1}(x)$ satisfies that $G_{x'}H$ contains only finitely many left cosets of H. In other words, $H \cap G_{x'}$ has finite index in $G_{x'}$. A fortiori, so does $G_{\{Y\}} \cap G_{x'}$ in $G_{x'}$. However, $G_{x'}/G_{\{Y\}} \cap G_{x'} \simeq G_{x'} \cdot Y$ which is finite if and only if $Y = \{x'\}$. Thus $H \leqslant G_{x'}$ and the index is finite. Note that $G_{x'} = G_{\{X'\}}$ where $X' = \mathrm{acl}_{\mathfrak{M}^{\mathrm{eq}}}(\emptyset)_{x'} \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$ which is autocommensurate by Lemma 3.8. Since H so is H, we have $H = G_{\{X'\}}$.
- Suppose that $\lambda_{G/H}$ is non-spherical. In this case, there exists $X \in \mathcal{C}(\mathbf{T}^{\mathrm{eq}})$, a finite subtree $X_0 \subseteq X$ and an irreducible representation of $G_{\{X\}}$ that is trivial on G_{X_0} such that $\lambda_{G/H}$ is isomorphic to $\pi := \mathrm{Ind}_{G_{\{X\}}}^{G}(\sigma)\colon G \curvearrowright \mathcal{H}$. Since $\lambda_{G/H}$ contains non-zero H-invariant vectors, so does π and we can fix such a vector $\xi \in \mathcal{H}\backslash\{0\}$. Let $g \in G$ be such that $\xi(g) \neq 0$. Since $\xi \in \ell^2(G/G_{\{X\}})$, the set $HgG_{\{X\}}$ contains only finitely many left $G_{\{X\}}$ cosets. In other words, $X' := g(X)$ satisfies that $H \cap G_{\{X'\}}$ has finite index in H. Let $h_1, \dots, h_n \in H$ be such that $H \subseteq \bigcup_i h_i H \cap G_{\{X'\}}$.

  Denoting $X_0' := g(X_0)$, we have $\mathcal{H}_{X_0'} = \pi(g)\mathcal{H}_{X_0} \neq 0$ hence there exists a $G_{X_0'}$-invariant function $f \in \ell^2(G/H)\backslash\{0\}$. Let $g' \in G$ be such that $f(g') \neq 0$. Then $G_{X_0'}g'H$ contains only finitely many H cosets. Since $G_{X_0'}$ has finite index in $G_{\{X'\}}$,

$$G_{\{X'\}}g'H \text{ also contains only finitely many } H\text{-cosets.} \tag{5}$$

  In other words, there exists $h_1', \dots, h_m' \in H$ such that $G_{\{X'\}}g'H \subseteq \bigcup_j h_i' H$. Consequently:

$$Hg'H \subseteq \bigcup_i h_i H \cap G_{\{X'\}}g'H \subseteq \bigcup_{i,j} h_i h_j' H.$$

  We have thus shown that $Hg'H$ contains only finitely many left cosets of H.

  We now show that $Hg'H$ contains only finitely many right cosets of H. Indeed, following the proof of [Tsa12, Lem 2.6], we define

$$\mathcal{F}(H) := \{uH\colon HuH \text{ contains only finitely many left } H\text{-cosets}\}.$$

  Suppose $u \in G$ is such that $uH \in \mathcal{F}(H)$. A fortiori, $G_Z u G_{\{Y\}}$ contains only finitely many left $G_{\{Y\}}$-cosets. In other words, $G_Z \cdot u(Y)$ is finite, i.e. $u(Y) \subseteq \mathrm{acl}_{\mathbf{T}^{\mathrm{eq}}}(Z)$ which is bounded. By Corollary 1.13, the set $E := \{u \in G\colon uH \in \mathcal{F}(H)\}$ is Roelcke precompact. Since H is open, there exists $F \subseteq E$ finite such that $E \subseteq HFH$. We now see that $\mathcal{F}(H)$ is finite. The proof of [Tsa12, Lem 2.6] now applies verbatim, showing that $Hg'H$ indeed contains only finitely many right cosets of H.

  We have thus shown that $g' \in \mathrm{Comm}_G(H) = H$. Then, (5) yields that $G_{\{X'\}} \cap H$ has finite index in $G_{\{X'\}}$. Therefore H and $G_{\{X'\}}$ are commensurate. Since both H and $G_{\{X'\}}$ autocommensurate, $H = G_{\{X\}}$.

□

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